Lesson 1 — Extra Practice

Powers, Exponent Vocabulary and the Product Law

Part A — Vocabulary, Brackets and Negative Bases

1. For each expression, state the **coefficient**, the **base** and the **exponent**. Read the brackets carefully — they decide what the base actually is.

(a) $-5x^4$ (b) $(-5x)^4$ (c) $7y$ (d) $-x^6$

2. Evaluate $3x^2$ and $(3x)^2$ when $x = 2$. Then explain, in a sentence, why the two expressions are *not* the same thing even though they use the same numbers.

3. Evaluate each product without a calculator. Expand first if it helps.

(a) $(-2)^3 \times (-2)^4$ (b) $-2^3 \times 2^4$ (c) $(-2)^2 \times (-2)^4$

(d) $-2^2 \times 2^4$ (e) $(-5) \times (-5)^2$ (f) -5×5^2

4. In question 3, parts (a) and (b) gave the same answer, and so did (e) and (f) — but (c) and (d) did not. Explain what the total number of negative factors has to do with this.

5. Amrit wants to use the Product Law on $(-2)^2 \times 2^3$ and writes $(-2)^5 = -32$. Evaluate the original expression yourself to show he is wrong, explain why the law does not apply as written, then rewrite $(-2)^2$ with the base 2 and finish the question correctly.

Part B — The Product Law with Coefficients and Several Variables

6. Simplify. Multiply the coefficients, then add the exponents of *each* variable separately.
(a) $(3x^2y)(4xy^5)(2x^3)$ (b) $(2a^3b)(5a^2b^4)(ab^2)$ (c) $(-3m^2n)(4m^5)(2n^3)$

7. Simplify. Watch the signs.
(a) $(5p^3q^2)(-2pq^3)$ (b) $(-c^4d)(-6c^2d^5)$ (c) $(4u^2v^3)(3u^5)(-v^2)$

8. Simplify each expression as far as the Product Law allows. Beside each answer, state whether the result **can** be written as a single power.
(a) $2^3 \times 5^2$ (b) $x^3y^2 \times x^2$ (c) $3^2 \times 3^3 \times 2^2$

9. **Error analysis.** Each student below made one mistake. Name the mistake, then give the correct answer.
(a) Jo writes $x^3 \cdot x^4 = x^{12}$. (b) Lena writes $3x^2 \cdot 4x^5 = 12x^{10}$. (c) Devin writes $2^3 \cdot 2^4 = 4^7$.

10. Simplify first, *then* substitute and evaluate. Substituting before simplifying is far more work.
(a) $2x^3 \cdot 3x^2$, $x = -2$ (b) $(-4a^2)(a^3)$, $a = 2$ (c) $(2m)(3m^2)(m^3)$, $m = -1$

Part C — Working Backwards and Rewriting Powers

11. Find the missing exponent in each statement.
(a) $x^n \cdot x^5 = x^{12}$ (b) $y^7 \cdot y^n = y^{20}$ (c) $m^3 \cdot m^n \cdot m^6 = m^{14}$ (d) $2^4 \cdot 2^n = 2^{10}$

12. These questions have more than one right answer.
(a) $(3x^a)(2x^b) = 6x^9$. Give **two** different pairs of whole numbers a and b that work, and state the rule every correct pair must follow.

(b) $(4y^m)(ky^n) = 28y^{10}$. Determine the value of k, then give **two** different pairs of whole numbers m and n that work.

13. Determine the missing coefficient k and the missing exponent n in each statement.
(a) $(kx^3)(5x^n) = 40x^{11}$ (b) $(ka^2b)(3ab^n) = 18a^3b^5$ (c) $(-2m^4)(km^n) = 14m^9$

14. A number can often be written as a power in more than one way — for example, $64 = 2^6 = 4^3 = 8^2$. Write each number below as a power in **every** way you can, using a whole-number base and an exponent greater than 1.
(a) 81 (b) 256 (c) 729 (d) 100

15. The Product Law needs *matching* bases. In each product below, rewrite both powers with the same base first — expanding into repeated multiplication is the safest way, since $4^3 = 4 \times 4 \times 4 = (2 \times 2)(2 \times 2)(2 \times 2) = 2^6$ — then apply the Product Law and give a single power.
(a) $4^3 \times 2^5$ (b) 9×27^2 (c) $8^2 \times 4^3$ (d) 25×5^3

Part D — Verifying, Applying and Stretching

16. **Does the law really work?** Evaluate each side of $2^3 \times 2^4 = 2^7$ separately and confirm the two values match. Then test whether $2^3 \times 3^4 = 6^7$ by evaluating both sides, and state what your result says about the Product Law.

17. A rectangle has width $3x^2$ and length $5x^4$. Write an expression for its area as a single term, then determine the area when $x = 2$.

18. A cube has edge length $2m^2$. Write an expression for the area of one face and one for the volume, then evaluate both when $m = 3$.

19. **Stretch.**
(a) The Product Law has a partner you will meet in Lesson 2: dividing powers with the same base *subtracts* the exponents, so $a^m \div a^n = a^{m-n}$. Use both laws to simplify $2^{10} \times 2^{15} \div 2^{20}$ mentally, then give the answer as a whole number.

(b) Which is larger, 2^{30} or 3^{20}? Do not use a calculator. *Hint:* 2^{30} is ten copies of 2^3 multiplied together, and 3^{20} is ten copies of 3^2 multiplied together. Explain your reasoning.

Answer Key. 1. (a) coefficient -5, base x, exponent 4 (b) coefficient 1, base $-5x$, exponent 4 (c) coefficient 7, base y, exponent 1 (d) coefficient -1, base x, exponent 6. **2.** $3x^2 = 3(2)^2 = 12$; $(3x)^2 = (6)^2 = 36$. In $3x^2$ only x is squared; the brackets in $(3x)^2$ make the whole product $3x$ the base, so the 3 gets squared too. **3.** (a) $(-2)^7 = -128$ (b) $-8 \times 16 = -128$ (c) $(-2)^6 = 64$ (d) $-4 \times 16 = -64$ (e) $(-5)^3 = -125$ (f) $-5 \times 25 = -125$. **4.** Each bracketed negative base contributes one negative factor. In (a)/(b) and (e)/(f) there is an odd number of negative factors either way, so both forms are negative and equal. In (c) there are six negative factors (even), giving a positive 64, while in (d) the single minus sign sits outside and stays, giving -64. **5.** $(-2)^2 \times 2^3 = 4 \times 8 = 32$, not -32. The bases are -2 and 2 — not identical — so the exponents cannot simply be added. Since the exponent 2 is even, $(-2)^2 = 4 = 2^2$, so $(-2)^2 \times 2^3 = 2^2 \times 2^3 = 2^5 = 32$. **6.** (a) $24x^6y^6$ (b) $10a^6b^7$ (c) $-24m^7n^4$. **7.** (a) $-10p^4q^5$ (b) $6c^6d^6$ (c) $-12u^7v^5$. **8.** (a) $8 \times 25 = 200$ — bases differ, so it cannot be written as a single power (b) x^5y^2 — two different variables, so no single power (c) $3^5 \times 2^2 = 243 \times 4 = 972$ — the 3s combine but the 2^2 cannot join them, so no single power. **9.** (a) She multiplied the exponents instead of adding them; $x^3 \cdot x^4 = x^7$. (b) She multiplied the exponents; the coefficients $3 \times 4 = 12$ were right, so $12x^7$. (c) He multiplied the bases as well as adding the exponents; the base stays the same, so $2^7 = 128$. **10.** (a) $6x^5$; $6(-2)^5 = 6(-32) = -192$ (b) $-4a^5$; $-4(2)^5 = -4(32) = -128$ (c) $6m^6$; $6(-1)^6 = 6(1) = 6$. **11.** (a) $n = 7$ (b) $n = 13$ (c) $n = 5$ (d) $n = 6$. **12.** (a) Any whole numbers with $a + b = 9$, e.g. $a = 4, b = 5$ or $a = 1, b = 8$; the coefficients already give $3 \times 2 = 6$. (b) $k = 7$, and any whole numbers with $m + n = 10$, e.g. $m = 3, n = 7$ or $m = 6, n = 4$. **13.** (a) $k = 8$, $n = 8$ (b) $k = 6$, $n = 4$ (c) $k = -7$, $n = 5$. **14.** (a) $81 = 3^4 = 9^2$ (b) $256 = 2^8 = 4^4 = 16^2$ (c) $729 = 3^6 = 9^3 = 27^2$ (d) $100 = 10^2$ only. **15.** (a) $2^6 \times 2^5 = 2^{11}$ (b) $3^2 \times 3^6 = 3^8$ (c) $2^6 \times 2^6 = 2^{12}$ (d) $5^2 \times 5^3 = 5^5$. **16.** $2^3 \times 2^4 = 8 \times 16 = 128$ and $2^7 = 128$ — they match. But $2^3 \times 3^4 = 8 \times 81 = 648$ while $6^7 = 279\,936$, so they do not match. The Product Law applies only when the bases are identical; you may never multiply the bases together. **17.** Area $= (3x^2)(5x^4) = 15x^6$; when $x = 2$, $15(64) = 960$ square units. **18.** Face area $= (2m^2)(2m^2) = 4m^4$; volume $= (4m^4)(2m^2) = 8m^6$. When $m = 3$, face area $= 4(81) = 324$ square units and volume $= 8(729) = 5832$ cubic units. **19.** (a) $2^{10} \times 2^{15} = 2^{25}$, then $2^{25} \div 2^{20} = 2^5 = 32$. (b) 3^{20} is larger. Group 2^{30} into ten blocks of $2^3 = 8$ and 3^{20} into ten blocks of $3^2 = 9$; both are a product of ten factors, but every factor of 3^{20} is 9 against a matching factor of 8, so $3^{20} > 2^{30}$. (Exactly, $3^{20} = 3\,486\,784\,401$ and $2^{30} = 1\,073\,741\,824$.)

Lesson 2 — Extra Practice

The Quotient Law and Power of a Power

Part A — The Quotient Law with Coefficients and Two Variables

When an expression has a coefficient and more than one variable, split the job into pieces: divide the coefficients, then apply the Quotient Law to *each* base separately. For example, $\dfrac{24x^7y^5}{6x^3y^2} = (24 \div 6)\left(x^{7-3}\right)\left(y^{5-2}\right) = 4x^4y^3$.

1. Simplify each expression. Keep all exponents positive.

(a) $\dfrac{45a^9b^6}{9a^4b^2}$ (b) $30m^8n^7 \div 5m^3n^4$ (c) $\dfrac{56p^{10}q^4}{8p^6q}$ (d) $72c^{11}d^9 \div 8c^5d^3$

2. Simplify. In each of these the coefficients do **not** divide evenly, so leave the number part as a reduced fraction.

(a) $\dfrac{15x^6}{6x^2}$ (b) $\dfrac{20a^7b^3}{8a^3b}$ (c) $18m^9n^5 \div 12m^5n^2$ (d) $\dfrac{42u^8v^6}{28u^3v^4}$

3. Simplify. Watch for three different bases, and for a variable that appears in only one part of the fraction.

(a) $\dfrac{36x^5y^8z^3}{9x^2y^3z}$ (b) $\dfrac{48a^7b^5}{6a^2}$ (c) $\dfrac{14r^6s^4t^9}{7r^4t^5}$

Part B — Power of a Power When a Coefficient Is Inside

An outer exponent applies to **everything** inside the brackets — the coefficient as well as each variable. Written out, $\left(2x^3\right)^4 = 2x^3 \times 2x^3 \times 2x^3 \times 2x^3 = 2^4x^{12} = 16x^{12}$. Forgetting to raise the coefficient is the most common slip in this lesson.

4. Simplify each expression. Raise the coefficient too.

(a) $\left(5a^4\right)^2$ (b) $\left(-3x^2y\right)^3$ (c) $\left(-2m^5\right)^3$ (d) $\left(4x^2y^3\right)^2$

5. Simplify. Pay close attention to the sign of each answer.

(a) $\left(-p^3\right)^5$ (b) $\left(-2c^4d\right)^4$ (c) $\left(3a^2b^5\right)^3$ (d) $\left(-k^6\right)^4$

6. The expressions $(3x)^2$ and $3x^2$ look almost the same but are not equal.

(a) Simplify $(3x)^2$.

(b) Evaluate both expressions when $x = 2$.

(c) Explain in one sentence what the brackets change.

Part C — Chains: Both Laws in One Expression

Work from the inside out. Clear every bracket first with the Power of a Power Law, then finish with the Quotient Law. For instance, $\dfrac{(2x^2)^3}{4x^4} = \dfrac{8x^6}{4x^4} = 2x^2$.

7. Simplify each expression.

(a) $\dfrac{(x^3)^4}{(x^2)^5}$

(b) $\dfrac{(a^5)^3}{(a^4)^2}$

(c) $\dfrac{(m^2)^6}{(m^3)^3}$

(d) $\dfrac{(y^4)^4}{(y^5)^2}$

8. Simplify. Remember that the outer exponent hits the coefficient as well.

(a) $\dfrac{(3y^4)^2}{9y^3}$

(b) $\dfrac{(5a^3)^2}{10a^2}$

(c) $\dfrac{(-3m^2)^3}{9m^4}$

9. Simplify. Both the numerator and the denominator need a bracket cleared first.

(a) $\dfrac{(2x^3)^4}{(4x^5)^2}$

(b) $\dfrac{(x^2y^3)^4}{(x^3y^5)^2}$

(c) $\dfrac{(2a^4b)^3}{(2a^2b)^2}$

Part D — Error Analysis

10. In each part, a student has made one mistake. Name the mistake, then give the correct simplified answer.

(a) Devon wrote $(x^3)^4 = x^7$.

(b) Priya wrote $(2x^3)^4 = 2x^{12}$.

(c) Malik wrote $\dfrac{12x^8}{3x^2} = 9x^6$.

(d) Jenna wrote $\dfrac{20a^6}{5a^2} = 4a^3$.

11. Sam claims that $\dfrac{x^{10}}{x^2} = x^5$ "because $10 \div 2 = 5$." Explain why the Quotient Law does not work that way, and support your explanation by writing $\dfrac{2^{10}}{2^2}$ as a single power and evaluating it both ways.

Part E — Working Backwards

12. Determine the missing whole-number exponent in each equation.

(a) $(x^a)^3 = x^{15}$ (b) $\dfrac{x^{12}}{x^{?}} = x^5$ (c) $\left(m^4\right)^n = m^{28}$ (d) $\dfrac{\left(y^3\right)^a}{y^4} = y^{11}$

13. Determine **both** missing values in each equation. *Hint:* start with the coefficient — it pins down the outer exponent first.

(a) $(3x^a)^b = 27x^{12}$ (b) $(2x^a)^b = 16x^{20}$ (c) $\left(5a^m b^2\right)^n = 125a^9b^6$

14. Fill in the missing expression so that each statement is true.

(a) $24x^9 \div$ _____ $= 4x^5$ (b) _____ $\div\, 7a^3 = 5a^4$ (c) (_____)$^3 = 8m^6n^9$

Part F — Numerical Checks, Reasoning, and Applications

15. Verify the Quotient Law numerically: show that $\dfrac{3^7}{3^4} = 3^3$ by evaluating the numerator, the denominator, and the quotient on the left side, and by evaluating 3^3 on the right side. State whether the two sides agree.

16. Verify the Power of a Power Law numerically: show that $\left(2^3\right)^4 = 2^{12}$ by evaluating each side. Then explain, using the meaning of an exponent, why the exponents get *multiplied* rather than added.

17. **A preview of Lesson 5.** Consider $\frac{x^5}{x^5}$, where $x \neq 0$.

(a) Simplify it using the Quotient Law.

(b) Simplify it by cancelling the five matching factors instead.

(c) What must x^0 therefore equal? Test your answer with $\frac{3^5}{3^5}$.

18. A large cube has an edge length of x^3 centimetres. A small cube has an edge length of x centimetres.

(a) Write the volume of each cube as a power.

(b) Write the ratio of the large volume to the small volume, simplified.

(c) How many small cubes fit in the large one when $x = 2$?

(d) Write the ratio of their surface areas, simplified.

19. A square has a side length of $4x^3$ metres and a rectangle has dimensions $2x^2$ metres by $8x^3$ metres.

(a) Write the area of the square as a simplified monomial.

(b) Write the area of the rectangle as a simplified monomial.

(c) Simplify the ratio of the square's area to the rectangle's area.

20. **Stretch.** Simplify $\frac{\left(2^3\right)^4 \times 2^5}{2^{12}}$ as a single power of 2, then evaluate it. Show every step: clear the bracket, use the Product Law on the numerator, then use the Quotient Law.

21. **Stretch.** Simplify $\frac{\left(3x^2y\right)^3 \times 2x^4}{\left(3x^3\right)^2 \times y}$ completely. Work out the numerator and the denominator separately before dividing.

22. **Stretch.** Without a calculator, decide which is larger: $\left(2^4\right)^3$ or $\left(2^3\right)^5$. Justify your choice by rewriting each as a single power of 2, and state how much larger it is as a power of 2.

Answer Key. 1. (a) $5a^5b^4$ (b) $6m^5n^3$ (c) $7p^4q^3$ (d) $9c^6d^6$. **2.** (a) $\frac{5}{2}x^4$ (b) $\frac{5}{2}a^4b^2$ (c) $\frac{3}{2}m^4n^3$ (d) $\frac{3}{2}u^5v^2$. **3.** (a) $4x^3y^5z^2$ (b) $8a^5b^5$ (c) $2r^2s^4t^4$. **4.** (a) $25a^8$ (b) $-27x^6y^3$ (c) $-8m^{15}$ (d) $16x^4y^6$. **5.** (a) $-p^{15}$ (b) $16c^{16}d^4$ (c) $27a^6b^{15}$ (d) k^{24}. **6.** (a) $9x^2$ (b) $(3x)^2 = 6^2 = 36$ and $3x^2 = 3(4) = 12$ (c) the brackets make the exponent apply to the 3 as well as the x; without them only the x is squared. **7.** (a) $\frac{x^{12}}{x^{10}} = x^2$ (b) $\frac{a^{15}}{a^8} = a^7$ (c) $\frac{m^{12}}{m^9} = m^3$ (d) $\frac{y^{16}}{y^{10}} = y^6$. **8.** (a) $\frac{9y^8}{9y^3} = y^5$ (b) $\frac{25a^6}{10a^2} = \frac{5}{2}a^4$ (c) $\frac{-27m^6}{9m^4} = -3m^2$. **9.** (a) $\frac{16x^{12}}{16x^{10}} = x^2$ (b) $\frac{x^8y^{12}}{x^6y^{10}} = x^2y^2$ (c) $\frac{8a^{12}b^3}{4a^4b^2} = 2a^8b$. **10.** (a) He added the exponents instead of multiplying them; $\left(x^3\right)^4 = x^{12}$. (b) She raised only the variable; the 2 must be raised too, giving $2^4x^{12} = 16x^{12}$. (c) He subtracted the coefficients instead of dividing them; $12 \div 3 = 4$, so the answer is $4x^6$. (d) She divided the exponents instead of subtracting them; $6 - 2 = 4$, so the answer is $4a^4$. **11.** The Quotient Law subtracts exponents, it does not divide them, so $\frac{x^{10}}{x^2} = x^{10-2} = x^8$. Check: $\frac{2^{10}}{2^2} = \frac{1024}{4} = 256 = 2^8$, while $2^5 = 32$, so Sam's rule fails. **12.** (a) $a = 5$ (b) $? = 7$ (c) $n = 7$ (d) $a = 5$. **13.** (a) $b = 3$, $a = 4$ (b) $b = 4$, $a = 5$ (c) $n = 3$, $m = 3$. **14.** (a) $6x^4$ (b) $35a^7$ (c) $2m^2n^3$. **15.** $3^7 = 2187$, $3^4 = 81$, $2187 \div 81 = 27$; $3^3 = 27$. The two sides agree. **16.** $2^3 = 8$ and $8^4 = 4096$; $2^{12} = 4096$, so the sides agree. The outer exponent says to write 4 copies of 2^3, which is 4 groups of 3 factors of 2 — that is $4 \times 3 = 12$ factors, so the exponents multiply. **17.** (a) $x^{5-5} = x^0$ (b) every factor cancels, leaving 1 (c) $x^0 = 1$; check $\frac{3^5}{3^5} = \frac{243}{243} = 1$. **18.** (a) large $= \left(x^3\right)^3 = x^9$, small $= x^3$ (b) $\frac{x^9}{x^3} = x^6$ (c) $2^6 = 64$ small cubes (d) $\frac{6\left(x^3\right)^2}{6x^2} = \frac{6x^6}{6x^2} = x^4$. **19.** (a) $\left(4x^3\right)^2 = 16x^6$ (b) $2x^2 \times 8x^3 = 16x^5$ (c) $\frac{16x^6}{16x^5} = x$ — the square is x times as large. **20.** $\left(2^3\right)^4 = 2^{12}$, so the numerator is $2^{12} \times 2^5 = 2^{17}$, and $\frac{2^{17}}{2^{12}} = 2^5 = 32$. **21.** Numerator: $27x^6y^3 \times 2x^4 = 54x^{10}y^3$. Denominator: $9x^6 \times y = 9x^6y$. Quotient: $\frac{54}{9}x^{10-6}y^{3-1} = 6x^4y^2$. **22.** $\left(2^4\right)^3 = 2^{12}$ and $\left(2^3\right)^5 = 2^{15}$, so $\left(2^3\right)^5$ is larger — by a factor of $2^3 = 8$ (32768 versus 4096).

Lesson 3 — Extra Practice

Power of a Product and Power of a Quotient

Part A — Every Factor Takes the Exponent

1. Each bracket holds **three or more factors**. Distribute the outer exponent onto every one of them — the coefficient included.
(a) $(2x^2y^3z)^4$ (b) $(3ab^4c^2)^3$ (c) $(5m^3np^2)^2$

2. Simplify. Beside each answer, write **even** or **odd** for the outside exponent, and circle the answers that came out negative.
(a) $(-3x^2)^3$ (b) $(-3x^2)^4$ (c) $(-2ab^3)^5$ (d) $(-4m^2n)^2$

3. These three expressions look almost the same but are not equal. Simplify each one, then evaluate all three at $x = 2$.
(a) $(-2x)^4$ (b) $-(2x)^4$ (c) $-2x^4$

4. Power of a quotient with a coefficient **top and bottom**. Raise every part of the numerator and every part of the denominator to the outer power.
(a) $\left(\dfrac{2x^3}{5y}\right)^3$ (b) $\left(\dfrac{-4a^2}{3b^3}\right)^2$ (c) $\left(\dfrac{3m^4}{2n}\right)^4$

5. Simplify each quotient. Watch the sign in part (a).
(a) $\left(\dfrac{-2p^3}{q^2}\right)^3$ (b) $\left(\dfrac{5x^2y}{4z^3}\right)^2$ (c) $\left(\dfrac{a^3b}{3c^2}\right)^3$

Part B — Nested Brackets and Mixed Laws

6. **Nested brackets.** Work from the inside out: simplify what is inside the round brackets first, then apply the outer exponent to that result.
(a) $\left[(2x^2)^3\right]^2$ (b) $\left[(-3y)^2\right]^3$ (c) $\left[\left(\frac{m^2}{2}\right)^2\right]^3$

7. Distribute each outer exponent first, then combine like bases with the Product or Quotient Law. Leave every answer with **positive exponents**. If a letter's bigger exponent is on the bottom, the leftover factors stay on the bottom: $\frac{x^2}{x^5} = \frac{1}{x^3}$.

(a) $\left(\frac{x^2}{y}\right)^3 \cdot (xy)^2$ (b) $(2a^3)^2 \cdot (3a)^2$ (c) $\frac{(3m^2n)^2}{(mn)^3}$

8. Simplify completely.

(a) $\frac{(4x^3)^2}{(2x)^3}$ (b) $(2xy^2)^3 \cdot \left(\frac{x}{y}\right)^2$ (c) $\left(\frac{a^4}{b^2}\right)^2 \div \left(\frac{a^3}{b}\right)^2$

9. **Stretch.** Simplify $\left(\frac{2x^2y}{4xy^3}\right)^3$ completely. Do it *twice*: first reduce the fraction inside the bracket and then cube it, and second cube everything first and reduce afterwards. Your two answers must match.

Part C — Finding the Mistake, Finding the Missing Piece

10. Each student below made the **same kind of error**: they forgot that something outside the variable also takes the exponent. Name the error and give the correct answer.

(a) Jaya wrote $(3x)^2 = 3x^2$. (b) Omar wrote $\left(\frac{x}{2}\right)^3 = \frac{x^3}{2}$.

11. Two more pieces of student work. Decide whether each is correct; if not, explain what went wrong and fix it.

(a) Priya wrote $(-5m^2)^2 = -25m^4$. (b) Devon wrote $(2a^3)^4 = 2a^{12}$.

12. **Work backwards.** Find the value of each unknown. Assume every unknown letter stands for a number.

(a) $(ax^b)^3 = 27x^{12}$; find a and b. (b) $\left(\frac{x^a}{3}\right)^2 = \frac{x^{10}}{9}$; find a. (c) $(2x^n)^4 = 16x^{20}$; find n.

13. Work backwards again. In part (b), give the **positive** value only.

(a) $(-2x^3)^n = 16x^{12}$; find n. (b) $\left(\dfrac{ay^2}{5}\right)^2 = \dfrac{49y^4}{25}$; find a. (c) $(cx^2y)^3 = -8x^6y^3$; find c.

Part D — Numerical Checks, Geometry, and Reasoning

14. A law is only trustworthy if the numbers agree.

(a) Evaluate $(2 \times 3)^3$ and $2^3 \times 3^3$ separately. Do they agree? Which law does this confirm?

(b) Now evaluate $(2+3)^3$ and 2^3+3^3 separately. Do they agree? What does this tell you about "distributing" an exponent over addition?

15. Repeat the same two tests with division and subtraction.

(a) Evaluate $\left(\frac{10}{5}\right)^3$ and $\dfrac{10^3}{5^3}$. Do they agree?

(b) Evaluate $(10-5)^3$ and 10^3-5^3. Do they agree? Write one sentence stating exactly which two operations an exponent may be distributed over.

16. A square has side length $3x^2$ centimetres.

(a) Write an expression for its area, and one for its perimeter.

(b) Evaluate both expressions when $x = 2$.

17. A cube has edge length $2xy$ metres.

(a) Write an expression for its volume and one for its total surface area (six square faces).

(b) Check your volume expression by letting $x = 1$ and $y = 2$: find the edge length, cube it, and compare with what your expression gives.

18. A cube has edge length e.

(a) Write its volume. Then write the volume of a cube whose edge is $2e$, and state how many times larger the second volume is.

(b) A square has side s. If the side is tripled to $3s$, how many times larger is the area? Explain why the multiplier is not 3.

19. *(Multiple Choice)* Which expression is equivalent to $\left(\dfrac{-2x^3}{y}\right)^3$?

A. $\dfrac{-6x^9}{y^3}$ B. $\dfrac{-8x^9}{y^3}$ C. $\dfrac{8x^6}{y^3}$ D. $\dfrac{-8x^{27}}{y^3}$

20. *(Numerical Response)* If $(2x^3y^2)^4 = kx^{12}y^8$, the value of k is _____.

21. Ravi says "an exponent spreads onto everything inside the brackets." Test his claim on $(x+y)^2$ and on $(xy)^2$ using $x = 3$ and $y = 4$, showing all four values. Then rewrite his statement so that it is true.

22. **Stretch.** Without a calculator, decide which of $(2^3)^4$, $(2 \times 3)^4$ and $\left(\frac{12}{2}\right)^4$ is largest. Show the single power of a number each one reduces to before you compare.

Answer Key. 1. (a) $16x^8y^{12}z^4$ (b) $27a^3b^{12}c^6$ (c) $25m^6n^2p^4$. **2.** (a) $-27x^6$, odd, negative (b) $81x^8$, even (c) $-32a^5b^{15}$, odd, negative (d) $16m^4n^2$, even. An even outside exponent always kills the negative sign. **3.** (a) $16x^4$; at $x = 2$, 256 (b) $-16x^4$; at $x = 2$, -256 (c) $-2x^4$; at $x = 2$, -32. **4.** (a) $\dfrac{8x^9}{125y^3}$ (b) $\dfrac{16a^4}{9b^6}$ (c) $\dfrac{81m^{16}}{16n^4}$. **5.** (a) $-\dfrac{8p^9}{q^6}$ (b) $\dfrac{25x^4y^2}{16z^6}$ (c) $\dfrac{a^9b^3}{27c^6}$. **6.** (a) $(8x^6)^2 = 64x^{12}$ (b) $(9y^2)^3 = 729y^6$ (c) $\left(\frac{m^4}{4}\right)^3 = \dfrac{m^{12}}{64}$. **7.** (a) $\dfrac{x^6}{y^3} \cdot x^2y^2 = \dfrac{x^8}{y}$ (b) $4a^6 \cdot 9a^2 = 36a^8$ (c) $\dfrac{9m^4n^2}{m^3n^3} = \dfrac{9m}{n}$. **8.** (a) $\dfrac{16x^6}{8x^3} = 2x^3$ (b) $8x^3y^6 \cdot \dfrac{x^2}{y^2} = 8x^5y^4$ (c) $\dfrac{a^8}{b^4} \div \dfrac{a^6}{b^2} = \dfrac{a^2}{b^2}$. **9.** Inside first: $\dfrac{2x^2y}{4xy^3} = \dfrac{x}{2y^2}$, so the cube is $\dfrac{x^3}{8y^6}$. Cubing first: $\dfrac{8x^6y^3}{64x^3y^9} = \dfrac{x^3}{8y^6}$ — the same. **10.** (a) She raised only x; the coefficient 3 must be squared too: $(3x)^2 = 9x^2$. (b) He raised only the numerator; the denominator gets the exponent as well: $\left(\frac{x}{2}\right)^3 = \dfrac{x^3}{8}$. **11.** (a) Incorrect. Squaring -5 gives $+25$, since an even power of a negative is positive: $(-5m^2)^2 = 25m^4$. (b) Incorrect. The coefficient 2 takes the exponent as well: $(2a^3)^4 = 16a^{12}$. **12.** (a) $a = 3$, $b = 4$ (b) $a = 5$ (c) $n = 5$. **13.** (a) $n = 4$ (b) $a = 7$ (c) $c = -2$. **14.** (a) $(2 \times 3)^3 = 6^3 = 216$ and $2^3 \times 3^3 = 8 \times 27 = 216$; they agree, confirming the Power of a Product Law. (b) $(2+3)^3 = 5^3 = 125$ but $2^3 + 3^3 = 8 + 27 = 35$; they do *not* agree — an exponent cannot be distributed over addition. **15.** (a) $\left(\frac{10}{5}\right)^3 = 2^3 = 8$ and $\dfrac{10^3}{5^3} = \dfrac{1000}{125} = 8$; they agree. (b) $(10-5)^3 = 125$ but $10^3 - 5^3 = 1000 - 125 = 875$; they do not agree. An exponent may be distributed over multiplication and division only, never over addition or subtraction. **16.** (a) Area $= (3x^2)^2 = 9x^4$ cm^2; perimeter $= 4(3x^2) = 12x^2$ cm. (b) At $x = 2$: area $= 9(16) = 144$ cm^2, perimeter $= 12(4) = 48$ cm. **17.** (a) Volume $= (2xy)^3 = 8x^3y^3$ m^3; surface area $= 6(2xy)^2 = 24x^2y^2$ m^2. (b) At $x = 1$, $y = 2$ the edge is 4 m and $4^3 = 64$; the expression gives $8(1)^3(2)^3 = 64$. They match. **18.** (a) e^3 and $(2e)^3 = 8e^3$; the volume is 8 times larger. (b) $(3s)^2 = 9s^2$, so the area is 9 times larger — the multiplier is squared along with the side, not just carried along. **19.** B. **20.** $k = 16$. **21.** $(3+4)^2 = 7^2 = 49$ but $3^2 + 4^2 = 9 + 16 = 25$; $(3 \times 4)^2 = 12^2 = 144$ and $3^2 \times 4^2 = 9 \times 16 = 144$. Ravi is wrong for addition. A true version: an exponent spreads onto every *factor* inside the brackets — that is, over multiplication and division only. **22.** $(2^3)^4 = 2^{12} = 4096$; $(2 \times 3)^4 = 6^4 = 1296$; $\left(\frac{12}{2}\right)^4 = 6^4 = 1296$. The largest is $(2^3)^4$, and the other two are equal.

Lesson 4 — Extra Practice

Combining Exponent Laws

Part A — Long Chains: Three or More Laws in One Expression

1. Each expression needs the power law *twice* before anything else can be combined. Clear both brackets, then finish.
(a) $(2x^3)^2 \times (3x^2)^3$ (b) $(a^2b^3)^2 \times (a^3b)^3$ (c) $(3m^2)^3 \times (2m^3)^2$

2. Clear the bracket, multiply the top together, then divide. Three laws each.
(a) $\dfrac{(2x^3y)^3 \times x^2}{4x^5y^2}$ (b) $\dfrac{(3a^2b)^2(2ab^3)}{6a^3b^4}$

3. Here the numerator **and** the denominator are each bracketed. Raise *both* coefficients before you divide anything.
(a) $\dfrac{(2x^2)^3}{(4x^3)^2}$ (b) $\dfrac{(4x^3)^2}{(2x^2)^3}$ (c) $\dfrac{(2a^2b^3)^4}{(4ab^2)^2}$

Compare your answers to (a) and (b). What do you notice, and why does it happen?

4. In each of these, **one of the two variables cancels completely**. Say which one, and give the simplified expression.
(a) $\dfrac{(2x^2y^3)^2}{4x^3y^6}$ (b) $\dfrac{(3a^2b^4)^2}{9ab^8}$ (c) $\dfrac{(2m^3n)^3}{8m^4n^3}$

5. Simplify. Each one mixes a bracket with both a multiplication and a division.
(a) $\dfrac{(2x^4)^3}{(2x^3)^2 \times x}$ (b) $(2a^2b)^3 \times \dfrac{3ab^2}{4a^4b^3}$

Part B — Order, Checking, and Error Analysis

6. **Does the order matter?** Consider $\left(\dfrac{12x^5y^2}{4x^2y}\right)^2$.

(a) Simplify *inside* the bracket first, then apply the outer power.

(b) Now square the numerator and the denominator first, then divide.

(c) Compare the two answers. Which route was less work, and why?

7. **Error analysis.** Bree simplified $\dfrac{(6x^4)^2 \times x^3}{(3x^2)^2}$ and wrote the work below. Exactly one step is wrong.

Step 1: $\dfrac{36x^8 \times x^3}{3x^4}$ Step 2: $\dfrac{36x^{11}}{3x^4}$ Step 3: $12x^7$

Identify the *first* incorrect step, explain in one sentence what she did wrong, and give the correct simplified form.

8. **The substitution check.** If two expressions are really equal, then substituting the same number into each must give the same value. Simplify each expression, then substitute $x = 2$ into the *original* and into your *simplified* form to confirm they agree.

(a) $\dfrac{(2x^3)^2 \times x}{4x^4}$

(b) $\dfrac{(3x^2)^2}{9x^3}$

9. Ali claims that $\dfrac{(2x^2)^3}{2x^3} = x^3$. Use the substitution check with $x = 2$ to show that his answer cannot be right, then give the correct simplified form.

10. True or false? If false, give the correct simplification.

(a) $\dfrac{(2x^3)^2}{2x^2} = x^4$

(b) $(x^2y)^3(xy^2)^2 = x^8y^7$

(c) $\left(\frac{6a^4}{3a^2}\right)^2 = 4a^4$

(d) $\dfrac{(4m^3)^2}{(2m^2)^2} = 2m^2$

Part C — Working Backwards, Applications, and Stretch

11. Find the missing number in each. Match the coefficients *and* the exponents.

(a) $\dfrac{(2x^a)^3}{x^4} = 8x^5$. Find a.

(b) $\dfrac{(3x^2)^n}{x^3} = 9x$. Find n.

(c) $(2x^3)^2 \times kx^4 = 20x^{10}$. Find k.

12. Find the missing *expression* (the one that belongs in the $\square$) that makes each statement true.

(a) $\dfrac{(3a^2b)^2}{\square} = 3ab$

(b) $\square \times (2x^2)^3 = 24x^{11}$

13. **Two cubes.** A large cube has edge length $2x^2$ and a small cube has edge length x (same units, $x > 0$).

(a) Write a simplified expression for each volume.

(b) Write the ratio of the large volume to the small volume, simplified.

(c) If $x = 3$, how many small cubes have the same total volume as one large cube?

14. A rectangle has area $(4x^3y^2)^2$ and width $8x^4y^3$. Determine a simplified expression for its length, then check your answer by multiplying the length by the width.

15. **Stretch.** Simplify. Handle the inside of the bracket, then the outer power, then the extra factor.

(a) $\left(\dfrac{3x^2y^3}{6xy}\right)^2 \cdot \dfrac{4x}{y^2}$

(b) $\left(\dfrac{4a^3b}{8ab^2}\right)^2 \times \dfrac{4b^3}{a}$

16. **Stretch (numbers only).** Simplify each to a single power, then evaluate it as a whole number.

(a) $\left(\dfrac{2^5 \times 2^3}{2^6}\right)^3 \times 2^2$

(b) $\dfrac{(3^2)^3 \times 3^4}{(3^3)^2}$

17. **Challenge.** Simplify $\dfrac{(2x^2y)^3 \times (3xy^2)^2}{(6x^2y^2)^2}$. Then explain, in one sentence, why it would be a poor idea to multiply out the numerator *before* clearing any of the brackets.

Answer Key. 1. (a) $4x^6 \times 27x^6 = 108x^{12}$ (b) $a^4b^6 \times a^9b^3 = a^{13}b^9$ (c) $27m^6 \times 4m^6 = 108m^{12}$. **2.** (a) $\dfrac{8x^9y^3 \times x^2}{4x^5y^2} = \dfrac{8x^{11}y^3}{4x^5y^2} = 2x^6y$ (b) $\dfrac{9a^4b^2 \times 2ab^3}{6a^3b^4} = \dfrac{18a^5b^5}{6a^3b^4} = 3a^2b$. **3.** (a) $\dfrac{8x^6}{16x^6} = \frac{1}{2}$ (b) $\dfrac{16x^6}{8x^6} = 2$ (c) $\dfrac{16a^8b^{12}}{16a^2b^4} = a^6b^8$. The answers to (a) and (b) are reciprocals, because the two expressions are upside-down versions of each other. **4.** (a) y cancels; $\dfrac{4x^4y^6}{4x^3y^6} = x$ (b) b cancels; $\dfrac{9a^4b^8}{9ab^8} = a^3$ (c) n cancels; $\dfrac{8m^9n^3}{8m^4n^3} = m^5$. **5.** (a) $\dfrac{8x^{12}}{4x^6 \times x} = \dfrac{8x^{12}}{4x^7} = 2x^5$ (b) $\dfrac{8a^6b^3 \times 3ab^2}{4a^4b^3} = \dfrac{24a^7b^5}{4a^4b^3} = 6a^3b^2$. **6.** (a) Inside gives $3x^3y$, and $(3x^3y)^2 = 9x^6y^2$. (b) $\dfrac{144x^{10}y^4}{16x^4y^2} = 9x^6y^2$. (c) Same answer both ways. Simplifying inside first is less work — the numbers stay small (3 and 12 instead of 144 and 16), and there is only one coefficient to square instead of two. **7. Step 1** is the first wrong step. She squared the coefficient in the numerator ($6 \to 36$) but not the one in the denominator: $(3x^2)^2 = 9x^4$, not $3x^4$. Correctly, $\dfrac{36x^8 \times x^3}{9x^4} = \dfrac{36x^{11}}{9x^4} = 4x^7$. **8.** (a) $\dfrac{4x^6 \times x}{4x^4} = \dfrac{4x^7}{4x^4} = x^3$. Check at $x = 2$: original $= \dfrac{16^2 \times 2}{4 \times 16} = \dfrac{512}{64} = 8$, and $2^3 = 8$. They agree. (b) $\dfrac{9x^4}{9x^3} = x$. Check at $x = 2$: original $= \dfrac{144}{72} = 2$, and $x = 2$. They agree. **9.** At $x = 2$ the original is $\dfrac{(2 \cdot 4)^3}{2 \cdot 8} = \dfrac{512}{16} = 32$, but Ali's $x^3 = 8$. Since $32 \neq 8$ his answer is wrong. Correctly $\dfrac{8x^6}{2x^3} = 4x^3$ (and at $x = 2$ that is 32). **10.** (a) False; $\dfrac{4x^6}{2x^2} = 2x^4$. (b) True; $x^6y^3 \times x^2y^4 = x^8y^7$. (c) True; inside is $2a^2$ and $(2a^2)^2 = 4a^4$. (d) False; $\dfrac{16m^6}{4m^4} = 4m^2$. **11.** (a) $\dfrac{8x^{3a}}{x^4} = 8x^{3a-4}$, so $3a - 4 = 5$ and $a = 3$. (b) $\dfrac{3^n x^{2n}}{x^3} = 9x$ needs $3^n = 9$ and $2n - 3 = 1$, so $n = 2$. (c) $4x^6 \times kx^4 = 4kx^{10} = 20x^{10}$, so $k = 5$. **12.** (a) $3a^3b$, since $\dfrac{9a^4b^2}{3a^3b} = 3ab$. (b) $3x^5$, since $3x^5 \times 8x^6 = 24x^{11}$. **13.** (a) Large: $(2x^2)^3 = 8x^6$; small: x^3. (b) $\dfrac{8x^6}{x^3} = 8x^3$. (c) $8(3)^3 = 8 \times 27 = 216$ small cubes. **14.** Length $= \dfrac{16x^6y^4}{8x^4y^3} = 2x^2y$. Check: $2x^2y \times 8x^4y^3 = 16x^6y^4 = (4x^3y^2)^2$. **15.** (a) Inside $= \frac{1}{2}xy^2$, squared $= \dfrac{x^2y^4}{4}$, times $\dfrac{4x}{y^2}$ gives x^3y^2. (b) Inside $= \dfrac{a^2}{2b}$, squared $= \dfrac{a^4}{4b^2}$, times $\dfrac{4b^3}{a}$ gives a^3b. **16.** (a) Inside $= 2^{8-6} = 2^2$, cubed $= 2^6$, times 2^2 gives $2^8 = 256$. (b) $\dfrac{3^6 \times 3^4}{3^6} = 3^4 = 81$. **17.** $\dfrac{8x^6y^3 \times 9x^2y^4}{36x^4y^4} = \dfrac{72x^8y^7}{36x^4y^4} = 2x^4y^3$. Multiplying first would build a large coefficient and long exponents before any cancelling; clearing the brackets first keeps every number small and lets matching bases divide out immediately.

Lesson 5 — Extra Practice

Zero Exponents

Part A — Where the Zero Exponent Attaches

The whole difficulty with this law is deciding *what the base is*. Brackets, coefficients, and negative signs all change the answer. In every question of Part A, assume each variable is non-zero.

1. Simplify each expression, where $x \neq 0$.

(a) $5x^0$ (b) $(5x)^0$ (c) $5\left(x^0\right)$ (d) $-5x^0$ (e) $(-5x)^0$

2. Simplify each expression, where $x \neq 0$ and $y \neq 0$.

(a) $\left(3x^2y\right)^0$ (b) $3\left(x^2y\right)^0$ (c) $3x^2y^0$ (d) $\left(3x^2\right)y^0$ (e) $(3x)^2y^0$

3. Evaluate each expression. Watch whether the negative sign is inside or outside the base.

(a) -4^0 (b) $(-4)^0$ (c) $-(-4)^0$ (d) $\left(-4^0\right)^2$ (e) $-\left(4^0\right)^3$

4. Exactly three of the five expressions below are equal to 1 when $m \neq 0$. State which three, and give the value of the other two.

$$7m^0, \qquad (7m)^0, \qquad 7^0m^0, \qquad \left(7m^0\right)^0, \qquad -(7m)^0$$

5. *(Error analysis)* Asked to simplify $5x^0$ for $x \neq 0$, a student wrote $5x^0 = 0$. Explain the mistake in words, then give the correct answer.

6. *(Error analysis)* Asked to simplify $(3x)^0$ for $x \neq 0$, a student wrote $(3x)^0 = 3$. Explain the mistake, give the correct answer, and write the expression that *does* simplify to 3.

Part B — Why $a^0 = 1$: Deriving and Justifying

7. *(Derive it.)* Let $x \neq 0$.

(a) Simplify $\dfrac{x^5}{x^5}$ using the Quotient Law, leaving your answer as a single power of x.

(b) Simplify $\dfrac{x^5}{x^5}$ a second way, without any exponent law, by thinking about what happens when a quantity is divided by itself. Then explain why (a) and (b) together force $x^0 = 1$.

8. *(Why the base must be non-zero.)* Repeat the argument of question 7 with $x = 0$: write out $\dfrac{0^5}{0^5}$ and explain exactly which step fails. Use this to explain why 0^0 is left undefined rather than being given the value 1.

9. State whether each claim is **true** or **false**, and justify your answer in one sentence. If a claim is false, give a specific counterexample.

(a) Any expression raised to the exponent 0 is equal to 1. (b) $x^0 = 1$ for every real number x.

(c) $-x^0 = (-x)^0$ for every $x \neq 0$.

(d) $\left(x^0\right)^5 = \left(x^5\right)^0$ for every $x \neq 0$.

(e) If $a \neq 0$ and $b \neq 0$, then $a^0 + b^0 = 2$.

(f) If $a \neq 0$, then $a^0 = a^1 \div a^1$.

10. *(Work backwards.)* Determine the missing value in each case. Assume every variable is non-zero.

(a) $\dfrac{x^7}{x^n} = 1$ (b) $\dfrac{\left(x^3\right)^4}{x^n} = 1$ (c) $\dfrac{x^{2n}}{x^{10}} = 1$ (d) $\dfrac{12b^n}{3b^4} = 4$

Part C — Zero Exponents Inside a Chain

11. Evaluate each expression. Assume every variable is non-zero.

(a) $2^0 + 3^0 + 4^0$ (b) $\left(3x^2\right)^0 + (4y)^0$ (c) $5^0 - 2^0 + (-3)^0$ (d) $2^0 + 2^1 + 2^2$

12. Simplify completely. Assume every variable is non-zero.

(a) $\dfrac{\left(2x^3\right)^0 \cdot 4x^2}{2x^2}$ (b) $\dfrac{\left(5a^4\right)^0 \cdot 9a^3}{3a}$ (c) $\left(4m^0\right)^2 \times m^3$

13. Each expression below collapses to a zero exponent. Simplify the numerator and denominator first, then finish. Assume every variable is non-zero.

(a) $\dfrac{x^4y^2}{x^4y^2}$ (b) $\dfrac{(2x)^3}{8x^3}$ (c) $\dfrac{\left(3a^2b\right)^2}{9a^4b^2}$

14. Simplify completely. Assume every variable is non-zero.

(a) $\dfrac{\left(2x^2y^0\right)^3}{4x^6}$ (b) $\dfrac{\left(6c^3\right)^0\left(2c^2\right)^3}{4c^4}$ (c) $\dfrac{15p^5q^0}{5p^5}$

15. Simplify completely. Assume every variable is non-zero.

(a) $\dfrac{\left(4a^2b^3\right)^0+(2a)^0}{(5b)^0}$ (b) $\left[\left(2x^3\right)^0+3x^0\right]^2$

Part D — Applications, Reasoning, and Stretch

16. A colony of bacteria is modelled by $P = 500 \cdot 2^t$, where P is the population and t is the time in hours since the colony was first measured.

(a) Determine P when $t = 0$.

(b) Explain why that value makes sense for this situation.

(c) Determine P when $t = 3$, and state how many times larger it is than the population at $t = 0$.

17. Evaluate $\left(7^0+7^1\right)^2$. A student claimed the answer was $7^0 + 7^2 = 50$ by "squaring each term." Show why that is wrong.

18. *(Stretch)* Simplify $\dfrac{\left(a^3b^2\right)^0(2ab)^2}{4a^2b^2}$, where $a \neq 0$ and $b \neq 0$, and state the single number it equals.

19. *(Stretch)* Simplify $\left[\dfrac{\left(3x^2\right)^0 \cdot 2x^5}{x^3}\right]^2$, where $x \neq 0$.

20. *(Multiple Choice)* If $x \neq 0$, which expression does **not** equal 1?

A. $(4x)^0$ B. $\dfrac{x^6}{x^6}$ C. $4x^0$ D. $\left(x^0\right)^4$

21. *(Multiple Choice)* For $x \neq 0$, the value of $\dfrac{(5x)^0 + 5x^0}{(2x)^0}$ is

A. 1 B. 5 C. 6 D. 10

22. *(Stretch — reasoning about the base.)* Remember that $a^0 = 1$ only when $a \neq 0$.

(a) For which real values of x is $(x-3)^0 = 1$? For which value does the expression fail to have a meaning, and why?

(b) For which real values of x is $\left(x^2-4\right)^0 = 1$? Justify your answer by stating which values of x must be excluded.

Answer Key. 1. (a) 5 (b) 1 (c) 5 (d) -5 (e) 1. **2.** (a) 1 (b) 3 (c) $3x^2$ (d) $3x^2$ (e) $9x^2$. **3.** (a) -1 (b) 1 (c) -1 (d) 1 (e) -1. **4.** Equal to 1: $(7m)^0$, 7^0m^0, and $\left(7m^0\right)^0$ (in the last one the bracket simplifies to 7 first, and $7^0 = 1$). The other two: $7m^0 = 7$ and $-(7m)^0 = -1$. **5.** The exponent 0 is attached only to x, not to the 5, and a zero exponent gives 1, not 0; so $5x^0 = 5 \times 1 = 5$. **6.** The brackets make the entire product $3x$ the base, so the whole thing becomes 1, not just the x part; $(3x)^0 = 1$. The expression equal to 3 is $3x^0$. **7.** (a) $\dfrac{x^5}{x^5} = x^{5-5} = x^0$. (b) Any non-zero quantity divided by itself is 1, so $\dfrac{x^5}{x^5} = 1$. The same expression cannot have two different values, so $x^0 = 1$. **8.** With $x = 0$, $\dfrac{0^5}{0^5} = \dfrac{0}{0}$, and the division-by-itself step fails because division by 0 is undefined. The argument that produced $a^0 = 1$ therefore never applies to $a = 0$, so 0^0 is left undefined. **9.** (a) F — the base must be non-zero; 0^0 is undefined. (b) F — it fails at $x = 0$. (c) F — e.g. at $x = 2$, $-x^0 = -1$ but $(-x)^0 = 1$. (d) T — both sides equal 1, since $\left(x^0\right)^5 = 1^5 = 1$ and $\left(x^5\right)^0 = 1$. (e) T — each power is 1, so the sum is 2. (f) T — $a^1 \div a^1 = a^{1-1} = a^0$, which is exactly the Quotient Law derivation. **10.** (a) $n = 7$ (b) $n = 12$ (c) $n = 5$ (d) $n = 4$. **11.** (a) 3 (b) 2 (c) 1 (d) 7. **12.** (a) 2 (b) $3a^2$ (c) $16m^3$. **13.** (a) 1 (b) $\dfrac{8x^3}{8x^3} = 1$ (c) $\dfrac{9a^4b^2}{9a^4b^2} = 1$. **14.** (a) 2 (b) $2c^2$ (c) 3. **15.** (a) $\dfrac{1+1}{1} = 2$ (b) $(1+3)^2 = 16$. **16.** (a) $P = 500 \cdot 2^0 = 500 \times 1 = 500$. (b) $t = 0$ is the moment of the first measurement, so no doubling has happened yet and P should be the starting population of 500 — the zero exponent gives exactly that. (c) $P = 500 \cdot 2^3 = 4000$, which is 8 times the starting population. **17.** $\left(7^0 + 7^1\right)^2 = (1+7)^2 = 8^2 = 64$. Squaring each term separately is not allowed because the brackets contain a *sum*, not a product: the Power of a Product Law applies to $(ab)^n$, not to $(a+b)^n$. (Indeed $64 \neq 50$.) **18.** $\dfrac{1 \cdot 4a^2b^2}{4a^2b^2} = 1$. **19.** $\left[\dfrac{2x^5}{x^3}\right]^2 = \left(2x^2\right)^2 = 4x^4$. **20.** C ($4x^0 = 4$). **21.** C. **22.** (a) All real x except $x = 3$; at $x = 3$ the base is 0 and 0^0 is undefined. (b) All real x except $x = 2$ and $x = -2$, since those are the values that make the base $x^2 - 4$ equal to 0.

Lesson 6 — Extra Practice

Negative Exponents

Part A — Where the Negative Attaches

A negative exponent belongs to *one base only* — the thing written immediately beneath it. Brackets change what that base is, and a negative *sign* out front is not the same thing as a negative *exponent*. In every question below, assume each variable is non-zero, and leave every answer **with positive exponents**.

1. Simplify each expression. Two of the five are equal — say which.

(a) $2x^{-3}$ (b) $(2x)^{-3}$ (c) $-2x^3$ (d) $(2x)^3$ (e) $2\left(x^{-3}\right)$

2. Evaluate each expression as an exact fraction. Watch the brackets.

(a) $3 \cdot 4^{-2}$ (b) $(3 \cdot 4)^{-2}$ (c) -4^{-2} (d) $(-4)^{-2}$ (e) $\left(\frac{1}{4}\right)^{-2}$

3. State whether each statement is True or False. If it is false, write the correct right-hand side with positive exponents.

(a) $(4x)^{-2} = \frac{1}{16x^2}$ (b) $4x^{-2} = \frac{1}{16x^2}$ (c) $(-6)^{-2} = \frac{1}{36}$ (d) $-6^{-2} = \frac{1}{36}$ (e) $\left(\frac{2}{5}\right)^{-1} = \frac{5}{2}$

4. Evaluate without a calculator. Give exact fractions, not decimals.

(a) 4^{-2} (b) $(-2)^{-3}$ (c) $\left(\frac{1}{5}\right)^{-2}$ (d) 10^{-3} (e) $(-1)^{-7}$

Part B — Moving Factors Across the Fraction Bar

Here is the shortcut that Part A has been building towards. A **factor** with a negative exponent may be moved to the other side of the fraction bar, and the exponent turns positive when it lands. For example $\dfrac{x^{-2}}{1} = \dfrac{1}{x^2}$ and $\dfrac{1}{y^{-4}} = \dfrac{y^4}{1}$. Notice what changes: the factor changes *level*, not *sign*. Nothing ever becomes negative, and a factor with an already-positive exponent stays exactly where it is. Leave every answer with positive exponents.

5. Rewrite each expression using positive exponents only.

(a) $\frac{3x^{-2}}{y^{-4}}$ (b) $\frac{a^{-5}}{b^3}$ (c) $\frac{m^4}{n^{-6}}$

6. *(Preview of Lesson 7)* Simplify each expression, leaving your answer with positive exponents.

(a) $\frac{5p^{-3}}{2q^{-1}}$ (b) $\frac{8c^{-2}d^5}{4c^3d^{-1}}$ (c) $\frac{x^{-3}y^2}{x^{-7}y^{-4}}$

7. *(Preview of Lesson 7)* Simplify each expression, leaving your answer with positive exponents. In (a) and (c) the *numbers* carry negative exponents too — move them the same way.

(a) $\frac{4^{-1}x^2}{3^{-2}x^5}$ (b) $\frac{6m^{-1}n^{-3}}{9m^{-4}n^2}$ (c) $\frac{(-3)^{-2}t^{-5}}{t^{-8}}$

8. A student writes $x^{-3} = -x^3$. Explain in one sentence what the student has confused, give the correct form, and then substitute $x = 2$ into both the student's answer and yours to show they are not equal.

9. A student writes $\dfrac{1}{x^{-2}} = \dfrac{1}{x^2}$. Explain what went wrong, give the correct form, and check your answer by substituting $x = 3$ into the original expression.

Part C — A Negative Exponent on a Fraction *(Preview of Lesson 7)*

If the whole base is a fraction, the negative exponent flips it. Since $\left(\dfrac{a}{b}\right)^{-n} = \dfrac{1}{(a/b)^n} = \dfrac{1}{a^n/b^n} = \dfrac{b^n}{a^n}$, the working rule is **take the reciprocal of the fraction, then raise it to the positive exponent**. Leave every answer with positive exponents.

10. Evaluate each expression as an exact fraction or whole number.

(a) $\left(\frac{2}{3}\right)^{-2}$ (b) $\left(\frac{3}{4}\right)^{-1}$ (c) $\left(\frac{1}{2}\right)^{-5}$ (d) $\left(-\frac{2}{5}\right)^{-3}$

11. Rewrite each expression with positive exponents.

(a) $\left(\frac{x}{y}\right)^{-3}$ (b) $\left(\frac{2a}{b^2}\right)^{-2}$ (c) $\left(\frac{3m^2}{n}\right)^{-1}$

12. **Justify the flip.** Starting from $\left(\dfrac{a}{b}\right)^{-3}$, apply the negative exponent rule first, then the power of a quotient law, and finally simplify the resulting complex fraction. Show every step and state the general result for $\left(\dfrac{a}{b}\right)^{-n}$.

Part D — Deriving the Rule and Working Backwards

13. **Where the rule comes from.** Simplify $\dfrac{x^3}{x^5}$ twice, in two different ways: once by subtracting the exponents with the quotient law, and once by writing out both powers in full and cancelling the common factors of x. Set the two results equal to each other and state what you have just proved about x^{-2}.

14. Determine the value of the unknown exponent in each equation.

(a) $2^n = \frac{1}{32}$ (b) $5^n = \frac{1}{125}$ (c) $x^{-a} = \frac{1}{x^7}$ (d) $10^n = 0.001$

15. Determine the value of the unknown exponent in each equation.

(a) $3^n = \frac{1}{81}$ (b) $\left(\frac{1}{2}\right)^n = 8$ (c) $4^n = \frac{1}{64}$ (d) $x^5 \cdot x^n = \frac{1}{x^2}$

16. Without a calculator, arrange the following from **least to greatest**. Write each one as an exact fraction first.

$$2^{-3}, \qquad 3^{-2}, \qquad \left(\tfrac{1}{2}\right)^{-1}, \qquad 5^{-1}, \qquad (-2)^{-3}$$

17. Without a calculator, arrange the following from **least to greatest**. Exactly two of them are equal — say which, and explain why.

$$4^{-1}, \qquad 2^{-2}, \qquad 8^{-1}, \qquad 3^{-1}, \qquad \left(\tfrac{1}{3}\right)^{-1}, \qquad 16^{-1}$$

Part E — Stretch

18. *(Preview of Lesson 7)* Simplify, leaving your answer with positive exponents. Work from the inside out: deal with the bracket first, then the division.

$$\frac{\left(2x^{-2}\right)^{-3}}{x^{-4}}$$

19. *(Preview of Lesson 7)* Simplify each expression, leaving your answer with positive exponents.

(a) $\left(\frac{3a^{-1}}{b^2}\right)^{-2}$ (b) $\frac{\left(x^{-3}y\right)^{-2}}{(xy^{-2})^3}$ (c) $\left(\frac{4m^{-2}}{m^5}\right)^{-1}$

20. Evaluate each expression as an exact fraction. In (c), notice that you must finish the addition inside the bracket *before* the outside exponent can be applied.

(a) $2^{-1} + 2^{-2} + 2^{-3}$ (b) $3^{-1} + 3^{-2}$ (c) $\left(2^{-1} + 3^{-1}\right)^{-1}$

21. Is it true that $(a+b)^{-1} = a^{-1} + b^{-1}$? Test it with $a = 2$ and $b = 3$, showing the exact value of both sides, and state your conclusion.

22. If $x^{-2} = \frac{1}{9}$, determine *all* possible values of x. Explain why there is more than one answer here but only one answer when $x^{-3} = \frac{1}{27}$.

Answer Key. 1. (a) $\frac{2}{x^3}$ (b) $\frac{1}{8x^3}$ (c) $-2x^3$ (d) $8x^3$ (e) $\frac{2}{x^3}$; (a) and (e) are equal, since brackets around x^{-3} alone change nothing. **2.** (a) $\frac{3}{16}$ (b) $\frac{1}{144}$ (c) $-\frac{1}{16}$ (d) $\frac{1}{16}$ (e) 16. **3.** (a) T (b) F — $4x^{-2} = \frac{4}{x^2}$; the 4 is a coefficient, not part of the base (c) T (d) F — $-6^{-2} = -\frac{1}{36}$; the sign is outside the power (e) T. **4.** (a) $\frac{1}{16}$ (b) $-\frac{1}{8}$ (c) 25 (d) $\frac{1}{1000}$ (e) -1. **5.** (a) $\frac{3y^4}{x^2}$ (b) $\frac{1}{a^5b^3}$ (c) m^4n^6. **6.** (a) $\frac{5q}{2p^3}$ (b) $\frac{2d^6}{c^5}$ (c) x^4y^6. **7.** (a) $\frac{9}{4x^3}$ (b) $\frac{2m^3}{3n^5}$ (c) $\frac{t^3}{9}$. **8.** The student turned the exponent's sign into the answer's sign; a negative exponent means *reciprocal*, so $x^{-3} = \frac{1}{x^3}$. At $x = 2$: $-x^3 = -8$ but $\frac{1}{x^3} = \frac{1}{8}$, and $-8 \neq \frac{1}{8}$. **9.** The student made the exponent positive but left the factor downstairs; a factor must change level as well, so $\frac{1}{x^{-2}} = x^2$. At $x = 3$: $\frac{1}{3^{-2}} = \frac{1}{1/9} = 9 = 3^2$. **10.** (a) $\frac{9}{4}$ (b) $\frac{4}{3}$ (c) 32 (d) $-\frac{125}{8}$. **11.** (a) $\frac{y^3}{x^3}$ (b) $\frac{b^4}{4a^2}$ (c) $\frac{n}{3m^2}$. **12.** $\left(\frac{a}{b}\right)^{-3} = \dfrac{1}{(a/b)^3} = \dfrac{1}{a^3/b^3} = 1 \times \dfrac{b^3}{a^3} = \dfrac{b^3}{a^3} = \left(\frac{b}{a}\right)^3$. In general $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$. **13.** Quotient law: $\dfrac{x^3}{x^5} = x^{3-5} = x^{-2}$. Cancelling: $\dfrac{x \cdot x \cdot x}{x \cdot x \cdot x \cdot x \cdot x} = \dfrac{1}{x \cdot x} = \dfrac{1}{x^2}$. The two results describe the same expression, so $x^{-2} = \dfrac{1}{x^2}$ — which is the negative exponent rule. **14.** (a) $n = -5$ (b) $n = -3$ (c) $a = 7$ (d) $n = -3$. **15.** (a) $n = -4$ (b) $n = -3$ (c) $n = -3$ (d) $n = -7$. **16.** $(-2)^{-3} = -\frac{1}{8}$, $3^{-2} = \frac{1}{9}$, $2^{-3} = \frac{1}{8}$, $5^{-1} = \frac{1}{5}$, $\left(\frac{1}{2}\right)^{-1} = 2$; so $(-2)^{-3} < 3^{-2} < 2^{-3} < 5^{-1} < \left(\frac{1}{2}\right)^{-1}$. **17.** $16^{-1} = \frac{1}{16}$, $8^{-1} = \frac{1}{8}$, $4^{-1} = 2^{-2} = \frac{1}{4}$, $3^{-1} = \frac{1}{3}$, $\left(\frac{1}{3}\right)^{-1} = 3$; so $16^{-1} < 8^{-1} < 4^{-1} = 2^{-2} < 3^{-1} < \left(\frac{1}{3}\right)^{-1}$. The equal pair is 4^{-1} and 2^{-2}, because $2^2 = 4$, so both are $\frac{1}{4}$. **18.** $\left(2x^{-2}\right)^{-3} = 2^{-3}x^6 = \dfrac{x^6}{8}$, then dividing by x^{-4} multiplies by x^4: $\dfrac{x^{10}}{8}$. **19.** (a) $\frac{a^2b^4}{9}$ (b) x^3y^4 (c) $\frac{m^7}{4}$. **20.** (a) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$ (b) $\frac{1}{3} + \frac{1}{9} = \frac{4}{9}$ (c) $\left(\frac{1}{2} + \frac{1}{3}\right)^{-1} = \left(\frac{5}{6}\right)^{-1} = \frac{6}{5}$. **21.** No. $(2+3)^{-1} = 5^{-1} = \frac{1}{5}$, while $2^{-1} + 3^{-1} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$, and $\frac{1}{5} \neq \frac{5}{6}$; an exponent cannot be distributed over a sum. **22.** $x^{-2} = \frac{1}{9}$ gives $x^2 = 9$, so $x = 3$ or $x = -3$ — an even power erases the sign, so both work. For $x^{-3} = \frac{1}{27}$ we get $x^3 = 27$, and an odd power keeps the sign, so only $x = 3$ works ($(-3)^3 = -27$).

Lesson 7 — Extra Practice

Combining Zero, Negative and Positive Exponents

Part A — Full Simplifications: Every Law in One Expression

Run the three-step checklist once, in order: brackets first, then combine matching bases, then clear the zero and negative exponents. Assume every variable is non-zero, and write **every final answer with positive exponents only**.

1. Simplify. Each expression mixes positive, negative and zero exponents.

(a) $\frac{\left(2x^{-2}y^{3}\right)^{-2}}{4x^{0}y^{-5}}$ (b) $\frac{\left(3a^{-1}b^{2}\right)^{3}}{9a^{2}b^{-1}}$ (c) $\left(2m^{0}n^{-3}\right)^{-2}\times 8n^{-1}$

2. Simplify. Here *both* the numerator and the denominator carry an outer power.

(a) $\frac{\left(4x^{3}y^{-1}\right)^{2}}{(2x^{-1}y^{2})^{3}}$ (b) $\frac{\left(5a^{2}b^{-3}\right)^{-1}}{(a^{-2}b)^{2}}$

3. Simplify. Take these one bracket at a time and keep the coefficients separate from the variables.

(a) $\frac{\left(3x^{-2}\right)^{-2}\times\left(2x^{3}y\right)^{2}}{6x^{0}y^{-1}}$ (b) $\frac{\left(2a^{3}b^{-1}\right)^{3}}{(4a^{2}b^{0})^{2}\times a^{-5}}$

4. Simplify. Watch for the zero exponents — decide carefully what each one is attached to.

(a) $\frac{x^{-3}y^{4}}{x^{2}y^{-1}}\times x^{0}y^{-2}$ (b) $\left(2p^{-1}q\right)^{-3}\times 4p^{0}q^{4}$ (c) $\frac{\left(m^{2}n^{-1}\right)^{-3}}{m^{-4}n^{0}}$

Part B — Brackets Raised to a Negative Power

A negative power on a bracket multiplies through to *every* factor inside — the coefficient included. On a fraction it also has a shortcut: $\left(\frac{a}{b}\right)^{-n}=\left(\frac{b}{a}\right)^{n}$. Simplifying inside the bracket before applying the outer power usually keeps the numbers small. Positive exponents in every final answer.

5. Simplify.

(a) $\left(\frac{2x^{3}}{y^{-2}}\right)^{-2}$ (b) $\left(\frac{a^{-2}}{3b}\right)^{-3}$ (c) $\left(\frac{4m^{0}}{n^{2}}\right)^{-1}$

6. Simplify inside the bracket first, then apply the outer power.

(a) $\left(\frac{6x^4y^{-1}}{2x^{-1}y^2}\right)^{-2}$ (b) $\left(\frac{5a^{-3}b^2}{10a^2b^{-1}}\right)^{-1}$

7. **Two routes, one answer.** Consider $\left(\frac{8x^5}{2x^2}\right)^{-2}$. Simplify it twice, by two different routes, and compare.

(a) **Route 1:** simplify inside the bracket first, then apply the outer -2.

(b) **Route 2:** apply the outer -2 to the numerator and to the denominator first, then simplify.

(c) Do the two routes agree? Which route involved smaller numbers, and why?

8. Simplify. Each of these has a bracket with a negative outer power *and* a factor sitting outside it.

(a) $\left(\frac{3x^{-1}y^2}{x^2y^{-2}}\right)^{-2} \times 9x^0y$ (b) $\left(\frac{2a^2}{b^{-3}}\right)^{-3} \times 4a^0b^9$

Part C — Collapsing, Checking, and Finding the Error

9. Each expression below collapses to a **constant** — every variable cancels out. Simplify each one fully and state the number that remains.

(a) $\frac{(2x)^0 \cdot 6x^2}{3x^2}$ (b) $\frac{\left(3a^{-2}\right)^2 \times a^4}{9a^0}$ (c) $\frac{\left(4y^3\right)^{-1} \times 8y^5}{2y^2}$

10. **Numerical spot-check.** Simplify $\frac{\left(2x^2y^{-1}\right)^2}{8x^0y^{-4}}$, then substitute $x = 2$ and $y = 1$ into *both* the original expression and your simplified form. The two values must match — show both calculations.

11. Kalvin simplifies $\dfrac{\left(3x^{-2}\right)^2}{9x^{-5}}$ and claims the answer is x^9. Substitute $x = 2$ into the original expression and into x^9 to show that his claim is wrong, then simplify the expression correctly.

12. **Error analysis.** Deven simplified $\dfrac{\left(2x^{-3}y\right)^{-2}}{4x^0y^{-3}}$ like this. Exactly *one* step is wrong.

$$\begin{aligned} \text{Step 1:} \quad & \left(2x^{-3}y\right)^{-2} = 2x^6y^{-2} \\ \text{Step 2:} \quad & \frac{2x^6y^{-2}}{4x^0y^{-3}} \\ \text{Step 3:} \quad & = \frac{x^6y}{2} \end{aligned}$$

Identify the incorrect step, explain in one sentence what Deven did wrong, then produce the correct simplified answer.

13. **Error analysis.** Priya simplified $\left(\dfrac{a^4}{2a^{-1}}\right)^{-2}$ like this. Again, exactly one step is wrong.

$$\begin{aligned} \text{Step 1:} \quad & \frac{a^4}{2a^{-1}} = \frac{a^5}{2} \\ \text{Step 2:} \quad & \left(\frac{a^5}{2}\right)^{-2} = \frac{a^{-10}}{2^{-2}} \\ \text{Step 3:} \quad & = \frac{1}{4a^{10}} \end{aligned}$$

Identify the incorrect step, explain the mistake, and give the correct answer.

Part D — Working Backwards, Ordering, and a Stretch

14. **Work backwards.** Determine the value of the unknown exponent in each equation.

(a) $\frac{x^a}{x^{-3}} = x^{10}$ (b) $\left(2x^{-a}\right)^{-2} = \frac{x^6}{4}$ (c) $\left(\frac{y^n}{y^2}\right)^{-3} = \frac{1}{y^9}$

15. **Work backwards with coefficients.** Determine the value of each unknown.

(a) $\left(ka^{-2}\right)^{-1} = \frac{a^2}{5}$

(b) $\frac{(3x^m)^2}{9x^{-4}} = x^{10}$

16. **Ordering.** Without a calculator, write each of the following as a fraction, then arrange them from **least to greatest**. Two of them are equal — say which two, and explain why.

$$2^{-3}, \qquad 3^{-2}, \qquad \left(\tfrac{1}{2}\right)^3, \qquad 4^{-1}$$

17. Compare these four values using exact fractions — no decimals, no calculator.

(a) Arrange from **greatest to least**:
3^{-1}, $\left(\frac{2}{3}\right)^{-2}$, 2^{-2}, $\left(\frac{1}{5}\right)^2$.

(b) Explain why a negative exponent on a number *greater* than 1 gives a result less than 1, while a negative exponent on a proper fraction gives a result greater than 1.

18. Evaluate without a calculator. Give an exact whole number or fraction.

(a) $\frac{2^{-3}\times 2^5}{2^0}$

(b) $\left(\frac{3^{-2}}{3^{-4}}\right)^{-1}$

(c) $\frac{\left(5^2\right)^{-1}\times 5^4}{5^0}$

19. **Stretch.** Simplify the following in three clear lines — brackets, then combine, then clear the negatives. The answer is clean.

$$\frac{\left(3x^{-1}y^3\right)^2 \times \left(2xy^0\right)^{-2}}{9x^0y^{-1}}$$

Then check your answer by substituting $x = 1$ and $y = 2$ into both the original expression and your simplified form.

20. *(Multiple Choice)* Choose the correct simplified form.

(a) $\frac{\left(3x^{-2}\right)^{-1}}{x^{-5}}$ is

A. $3x^7$ B. $\frac{x^7}{3}$ C. $\frac{3}{x^7}$ D. $\frac{1}{3x^3}$

(b) $\left(\frac{2}{x^{-1}}\right)^{-3}$ is

A. $\frac{1}{8x^3}$ B. $8x^3$ C. $\frac{x^3}{8}$ D. $-8x^3$

21. Using only the definition $a^{-n} = \dfrac{1}{a^n}$ and the law $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$, explain why $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$. Then use that result to simplify $\left(\dfrac{3x^2}{4y}\right)^{-2}$ in a single step.

22. **Stretch — find the missing value.**

(a) Determine the positive value of k for which $\dfrac{(kx^{-2})^2}{8x^{-7}}$ simplifies to $2x^3$. (b) Determine the value of n for which $\left(\frac{x^n}{x^{-2}}\right)^{-3} = \frac{1}{x^{12}}$.

Answer Key. 1. (a) $\dfrac{x^4}{16y}$ (b) $\dfrac{3b^7}{a^5}$ (c) $2n^5$. **2.** (a) $\dfrac{2x^9}{y^8}$ (b) $\dfrac{a^2b}{5}$. **3.** (a) $(3x^{-2})^{-2} = \dfrac{x^4}{9}$ and $(2x^3y)^2 = 4x^6y^2$, giving $\dfrac{2x^{10}y^3}{27}$ (b) $\dfrac{a^{10}}{2b^3}$. **4.** (a) $\dfrac{y^3}{x^5}$ (b) $\dfrac{p^3q}{2}$ (c) $\dfrac{n^3}{m^2}$. **5.** (a) $\dfrac{1}{4x^6y^4}$ (b) $27a^6b^3$ (c) $\dfrac{n^2}{4}$. **6.** (a) inside $= 3x^5y^{-3}$, so $\dfrac{y^6}{9x^{10}}$ (b) inside $= \frac{1}{2}a^{-5}b^3$, so $\dfrac{2a^5}{b^3}$. **7.** (a) inside $= 4x^3$, so $(4x^3)^{-2} = \dfrac{1}{16x^6}$ (b) $\dfrac{(8x^5)^{-2}}{(2x^2)^{-2}} = \dfrac{4x^4}{64x^{10}} = \dfrac{1}{16x^6}$ (c) Yes, both give $\dfrac{1}{16x^6}$; Route 1 kept the numbers smaller because the coefficient $8 \div 2 = 4$ was reduced *before* being squared, instead of squaring 8 and 2 separately. **8.** (a) inside $= 3x^{-3}y^4$; $(3x^{-3}y^4)^{-2} = \dfrac{x^6}{9y^8}$, times $9y$ gives $\dfrac{x^6}{y^7}$ (b) $(2a^2b^3)^{-3} = \dfrac{1}{8a^6b^9}$, times $4b^9$ gives $\dfrac{1}{2a^6}$. **9.** (a) $\dfrac{6x^2}{3x^2} = 2$ (b) $\dfrac{9a^{-4}\cdot a^4}{9} = 1$ (c) $\dfrac{2y^2}{2y^2} = 1$. **10.** $\dfrac{4x^4y^{-2}}{8y^{-4}} = \dfrac{x^4y^2}{2}$. Check: original $= \dfrac{(2\cdot 4\cdot 1)^2}{8\cdot 1} = \dfrac{64}{8} = 8$; simplified $= \dfrac{16\cdot 1}{2} = 8$. They agree. **11.** At $x = 2$ the original is $\dfrac{9/16}{9/32} = 2$, but $2^9 = 512$, so x^9 is wrong. Correctly, $\dfrac{9x^{-4}}{9x^{-5}} = x^{-4-(-5)} = x$ (and $x = 2$ gives 2). **12.** **Step 1** is wrong: the outer -2 must be applied to the coefficient 2 as well, so $(2x^{-3}y)^{-2} = 2^{-2}x^6y^{-2} = \dfrac{x^6}{4y^2}$, not $2x^6y^{-2}$. Steps 2 and 3 are then correct in method. Correct answer: $\dfrac{x^6/(4y^2)}{4y^{-3}} = \dfrac{x^6y}{16}$. **13.** **Step 3** is wrong: 2^{-2} in the *denominator* means dividing by $\frac{1}{4}$, which multiplies by 4. So $\dfrac{a^{-10}}{2^{-2}} = \dfrac{4}{a^{10}}$, not $\dfrac{1}{4a^{10}}$. Steps 1 and 2 are correct. Correct answer: $\dfrac{4}{a^{10}}$. **14.** (a) $a - (-3) = 10$, so $a = 7$ (b) $(2x^{-a})^{-2} = \dfrac{x^{2a}}{4}$, so $2a = 6$ and $a = 3$ (c) $(y^{n-2})^{-3} = y^{6-3n} = y^{-9}$, so $n = 5$. **15.** (a) $(ka^{-2})^{-1} = \dfrac{a^2}{k}$, so $k = 5$ (b) $\dfrac{9x^{2m}}{9x^{-4}} = x^{2m+4} = x^{10}$, so $m = 3$. **16.** $2^{-3} = \frac{1}{8}$, $3^{-2} = \frac{1}{9}$, $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, $4^{-1} = \frac{1}{4}$. Least to greatest: $3^{-2} < 2^{-3} = \left(\frac{1}{2}\right)^3 < 4^{-1}$. The equal pair is 2^{-3} and $\left(\frac{1}{2}\right)^3$, because $2^{-3} = \dfrac{1}{2^3} = \left(\frac{1}{2}\right)^3$ — a negative exponent is exactly the same as cubing the reciprocal. **17.** (a) $\left(\frac{2}{3}\right)^{-2} = \frac{9}{4}$, $3^{-1} = \frac{1}{3}$, $2^{-2} = \frac{1}{4}$, $\left(\frac{1}{5}\right)^2 = \frac{1}{25}$, so $\left(\frac{2}{3}\right)^{-2} > 3^{-1} > 2^{-2} > \left(\frac{1}{5}\right)^2$. (b) A negative exponent gives the reciprocal of a power. If the base is greater than 1, that power is greater than 1 and its reciprocal is less than 1; if the base is a proper fraction, the power is even smaller and its reciprocal is greater than 1. **18.** (a) $2^2 = 4$ (b) $\left(3^2\right)^{-1} = \frac{1}{9}$ (c) $5^{-2} \times 5^4 = 25$. **19.** $(3x^{-1}y^3)^2 = 9x^{-2}y^6$ and $(2xy^0)^{-2} = \dfrac{1}{4x^2}$, so the numerator is $\dfrac{9y^6}{4x^4}$; dividing by $9y^{-1}$ gives $\dfrac{y^7}{4x^4}$. Check at $x = 1$, $y = 2$: original $= \dfrac{24^2 \times \frac{1}{4}}{9 \times \frac{1}{2}} = \dfrac{144}{4.5} = 32$, and $\dfrac{2^7}{4} = \dfrac{128}{4} = 32$. **20.** (a) B, since $(3x^{-2})^{-1} = \frac{x^2}{3}$ and dividing by x^{-5} gives $\frac{x^7}{3}$ (b) A, since $\frac{2}{x^{-1}} = 2x$ and $(2x)^{-3} = \frac{1}{8x^3}$. **21.** $\left(\dfrac{a}{b}\right)^{-n} = \dfrac{1}{(a/b)^n} = \dfrac{1}{a^n/b^n} = \dfrac{b^n}{a^n} = \left(\dfrac{b}{a}\right)^n$. Applying it, $\left(\dfrac{3x^2}{4y}\right)^{-2} = \left(\dfrac{4y}{3x^2}\right)^2 = \dfrac{16y^2}{9x^4}$. **22.** (a) $\dfrac{k^2x^{-4}}{8x^{-7}} = \dfrac{k^2x^3}{8} = 2x^3$ requires $k^2 = 16$, so $k = 4$ (b) $\left(x^{n+2}\right)^{-3} = x^{-3n-6} = x^{-12}$, so $3n + 6 = 12$ and $n = 2$.

Lesson 8 — Extra Practice

Scientific Notation

Part A — Conversions, and Answers That Are Not Finished

Throughout this set, an answer is only in scientific notation when it has the form $a \times 10^n$ with n an integer and the coefficient in the range $1 \le a < 10$. A single nonzero digit sits in front of the decimal point — no more, no less.

1. Write each number in scientific notation.
(a) 0.000000705 (b) 12 340 000 (c) 6 020 000 000 (d) 0.0000000009 (e) 80.5

2. Write each number in standard notation. Watch the placeholder zeros in (c).
(a) 3.06×10^{-5} (b) 7.2×10^{7} (c) 5.004×10^{-3} (d) 1.1×10^{-6}

3. None of the numbers below is written correctly in scientific notation. For each one, state what is wrong with the coefficient, then rewrite the number correctly. *Hint:* the value must not change — if you shrink the coefficient, the exponent must grow.
(a) 45×10^{6} (b) 0.62×10^{-3} (c) 18.4×10^{-5} (d) 0.05×10^{8}

Part B — Multiplying and Dividing *(Preview of Lesson 10)*

Multiply the coefficients and add the exponents; divide the coefficients and subtract the exponents. The extra step — the one that is easy to forget — is that the result is often *not yet* in scientific notation:

$$\left(6 \times 10^{4}\right)\left(5 \times 10^{7}\right) = 30 \times 10^{11} = 3.0 \times 10^{1} \times 10^{11} = 3 \times 10^{12}.$$

Every answer in this Part must end with a coefficient in $[1, 10)$.

4. Multiply. Write each answer in scientific notation.
(a) $\left(2 \times 10^{5}\right)\left(3 \times 10^{8}\right)$ (b) $\left(4 \times 10^{6}\right)\left(2.5 \times 10^{3}\right)$ (c) $\left(3 \times 10^{-4}\right)\left(2 \times 10^{-7}\right)$ (d) $\left(5 \times 10^{9}\right)\left(8 \times 10^{-4}\right)$

5. Multiply. **Every part here needs renormalising** — the first coefficient you get will be 10 or larger.
(a) $\left(7 \times 10^{3}\right)\left(8 \times 10^{5}\right)$ (b) $\left(9 \times 10^{-6}\right)\left(4 \times 10^{-2}\right)$ (c) $\left(2.5 \times 10^{7}\right)\left(6 \times 10^{-3}\right)$ (d) $\left(8.4 \times 10^{-5}\right)\left(5 \times 10^{9}\right)$

6. Divide. In parts (b) and (c) the coefficient comes out *less* than 1, so the exponent must go **down** by one.
(a) $\dfrac{8 \times 10^{9}}{2 \times 10^{4}}$ (b) $\dfrac{3 \times 10^{6}}{6 \times 10^{-2}}$ (c) $\dfrac{1.2 \times 10^{-3}}{4 \times 10^{5}}$ (d) $\dfrac{9.6 \times 10^{8}}{1.6 \times 10^{-3}}$

7. Evaluate each power. Apply the exponent to *both* the coefficient and the power of ten, then renormalise.
(a) $\left(3 \times 10^{-4}\right)^{3}$ (b) $\left(2 \times 10^{5}\right)^{4}$ (c) $\left(5 \times 10^{-3}\right)^{2}$

8. **Comparing magnitudes.** "How many times larger?" is a division question. Divide the larger number by the smaller one and leave the answer in scientific notation.
(a) How many times larger is 6×10^{8} than 2×10^{5}? (b) How many times larger is 9×10^{-2} than 3×10^{-7}? (c) How many times larger is 1.5×10^{11} than 5×10^{6}?

Part C — Adding and Subtracting *(Preview of Lesson 10)*

Adding is *harder* than multiplying, because you cannot add the exponents — the two numbers must first be rewritten with the **same** power of ten. Rewrite the smaller-exponent number to match the larger one, add or subtract the coefficients, then renormalise if needed:

$$3.2 \times 10^{5} + 4 \times 10^{4} \;=\; 3.2 \times 10^{5} + 0.4 \times 10^{5} \;=\; 3.6 \times 10^{5}.$$

Notice the 0.4: it is deliberately outside $[1, 10)$ in the middle of the work, and that is fine. Only the *final* answer must be in proper scientific notation.

9. Add or subtract. The exponents already match — but check your final coefficient in part (c).
(a) $4.2 \times 10^{6} + 3.5 \times 10^{6}$ (b) $8.1 \times 10^{-4} - 2.6 \times 10^{-4}$ (c) $6.5 \times 10^{3} + 5.8 \times 10^{3}$

10. Add or subtract. Match the exponents first.
(a) $5 \times 10^{7} + 3 \times 10^{6}$ (b) $7.2 \times 10^{-3} - 4 \times 10^{-4}$ (c) $9 \times 10^{5} + 2.5 \times 10^{4}$ (d) $1.4 \times 10^{-2} + 6 \times 10^{-3}$

11. Add or subtract. In part (b) the subtraction leaves a coefficient smaller than 1, so the exponent must drop.
(a) $3 \times 10^{8} - 4 \times 10^{7}$ (b) $1.02 \times 10^{-5} - 9 \times 10^{-6}$ (c) $2 \times 10^{4} + 7 \times 10^{3} + 5 \times 10^{2}$

Part D — Ordering and Working Backwards

12. Order each list from **smallest to largest**. Compare exponents first; compare coefficients only when the exponents are equal.
(a) 5×10^{-3}, 2×10^{4}, 8×10^{-5}, 1.5×10^{-3}, 9×10^{3}, 4×10^{-4}

(b) 6.2×10^{-6}, 6.2×10^{-5}, 5.9×10^{-5}, 7×10^{-6}

13. Determine the missing value in each statement.
(a) $n \times 10^5 = 340\,000$ (b) $7.1 \times 10^n = 0.00071$ (c) $n \times 10^{-3} = 0.0092$ (d) $2.5 \times 10^n = 2\,500\,000$

14. *(Preview of Lesson 10)* **What happens when you double?** Suppose a number is written as $a \times 10^b$ in proper scientific notation.
(a) Double 3×10^7 and write the result in scientific notation. What changed?
(b) Double 6×10^7 and write the result in scientific notation. What changed this time?
(c) Using (a) and (b), state exactly which values of a make doubling increase the exponent. Explain.

Part E — Real-World Data *(Preview of Lesson 10)*

15. Light travels at about 3×10^5 km/s, and the Sun is about 1.5×10^8 km from Earth.
(a) How many seconds does sunlight take to reach Earth? Give the answer in scientific notation.
(b) *(calculator)* Convert your answer to minutes, rounded to one decimal place.

16. A proton has a mass of about 1.67×10^{-27} kg and an electron has a mass of about 9.1×10^{-31} kg.
(a) Determine the total mass of 3×10^8 protons, in scientific notation.
(b) *(calculator)* How many times more massive is a proton than an electron? Round your coefficient to one decimal place.

17. *(calculator)* A bacterial culture starts with 5×10^3 cells, and the number of cells doubles once every 20 minutes. Determine the number of cells after 200 minutes, and write it in scientific notation. *Hint:* first work out how many doublings occur.

18. **Unit conversion required.** Answer each question in scientific notation.
(a) Red light has a wavelength of about 700 nanometres. Given that 1 nm = 1×10^{-9} m, express this wavelength in metres.
(b) A country's national debt is about 1.2 trillion dollars. Write this amount in dollars, in scientific notation. *(One trillion is* 10^{12}*.)*
(c) If that country has a population of 4×10^7 people, what is the debt per person?

Part F — Error Analysis and Stretch *(Preview of Lesson 10)*

19. Each student below made a different mistake. Identify the error in one sentence, then give the correct answer.

(a) Devon computed $(4 \times 10^5)(3 \times 10^6)$ and wrote 12×10^{11} as his final answer in scientific notation.

(b) Priya wrote 2.6×10^{-4} in standard notation as 26 000, saying she moved the decimal 4 places.

(c) Malik computed $5 \times 10^3 + 3 \times 10^2$ by adding the coefficients and the exponents, getting 8×10^5.

(d) Sara simplified $(2 \times 10^{-3})^2$ to 2×10^{-6}, squaring the power of ten but leaving the coefficient alone.

20. **Stretch.** Astronomers measure time in enormous units, so scientific notation is the only practical way to write it.

(a) *(calculator)* Using a 365-day year, determine the number of seconds in one year. Write the exact value in standard notation and then in scientific notation.

(b) The universe is about 1.4×10^{10} years old. Using the rounded value 3.15×10^7 seconds per year, estimate its age in seconds, in scientific notation.

Answer Key. 1. (a) 7.05×10^{-7} (b) 1.234×10^7 (c) 6.02×10^9 (d) 9×10^{-10} (e) 8.05×10^1. **2.** (a) 0.0000306 (b) 72 000 000 (c) 0.005004 (d) 0.0000011. **3.** (a) $45 \geq 10$; 4.5×10^7 (b) $0.62 < 1$; 6.2×10^{-4} (c) $18.4 \geq 10$; 1.84×10^{-4} (d) $0.05 < 1$; 5×10^6. **4.** (a) 6×10^{13} (b) $10 \times 10^9 = 1 \times 10^{10}$ (c) 6×10^{-11} (d) $40 \times 10^5 = 4 \times 10^6$. **5.** (a) $56 \times 10^8 = 5.6 \times 10^9$ (b) $36 \times 10^{-8} = 3.6 \times 10^{-7}$ (c) $15 \times 10^4 = 1.5 \times 10^5$ (d) $42 \times 10^4 = 4.2 \times 10^5$. **6.** (a) 4×10^5 (b) $0.5 \times 10^8 = 5 \times 10^7$ (c) $0.3 \times 10^{-8} = 3 \times 10^{-9}$ (d) 6×10^{11}. **7.** (a) $27 \times 10^{-12} = 2.7 \times 10^{-11}$ (b) $16 \times 10^{20} = 1.6 \times 10^{21}$ (c) $25 \times 10^{-6} = 2.5 \times 10^{-5}$. **8.** (a) 3×10^3 (3000 times) (b) 3×10^5 (c) 3×10^4. **9.** (a) 7.7×10^6 (b) 5.5×10^{-4} (c) $12.3 \times 10^3 = 1.23 \times 10^4$. **10.** (a) $5 \times 10^7 + 0.3 \times 10^7 = 5.3 \times 10^7$ (b) $7.2 \times 10^{-3} - 0.4 \times 10^{-3} = 6.8 \times 10^{-3}$ (c) $9 \times 10^5 + 0.25 \times 10^5 = 9.25 \times 10^5$ (d) $1.4 \times 10^{-2} + 0.6 \times 10^{-2} = 2 \times 10^{-2}$. **11.** (a) $3 \times 10^8 - 0.4 \times 10^8 = 2.6 \times 10^8$ (b) $1.02 \times 10^{-5} - 0.9 \times 10^{-5} = 0.12 \times 10^{-5} = 1.2 \times 10^{-6}$ (c) $20\,000 + 7\,000 + 500 = 27\,500 = 2.75 \times 10^4$. **12.** (a) $8 \times 10^{-5} < 4 \times 10^{-4} < 1.5 \times 10^{-3} < 5 \times 10^{-3} < 9 \times 10^3 < 2 \times 10^4$ (b) $6.2 \times 10^{-6} < 7 \times 10^{-6} < 5.9 \times 10^{-5} < 6.2 \times 10^{-5}$. **13.** (a) $n = 3.4$ (b) $n = -4$ (c) $n = 9.2$ (d) $n = 6$. **14.** (a) 6×10^7 — the coefficient doubles, the exponent is unchanged. (b) $12 \times 10^7 = 1.2 \times 10^8$ — the coefficient doubled past 10, so renormalising raised the exponent by 1. (c) The exponent increases exactly when $2a \geq 10$, that is when $a \geq 5$; for $1 \leq a < 5$ the doubled coefficient stays below 10 and the exponent does not change. **15.** (a) $(1.5 \times 10^8) \div (3 \times 10^5) = 0.5 \times 10^3 = 5 \times 10^2$ s (that is 500 s) (b) $500 \div 60 \approx 8.3$ minutes. **16.** (a) $(1.67 \times 10^{-27})(3 \times 10^8) = 5.01 \times 10^{-19}$ kg (b) $(1.67 \times 10^{-27}) \div (9.1 \times 10^{-31}) \approx 0.18 \times 10^4 \approx 1.8 \times 10^3$, so a proton is about 1800 times more massive. **17.** $200 \div 20 = 10$ doublings, so $(5 \times 10^3)(2^{10}) = (5 \times 10^3)(1024) = 5\,120\,000 = 5.12 \times 10^6$ cells. **18.** (a) $700 \times (1 \times 10^{-9}) = 7 \times 10^{-7}$ m (b) 1.2×10^{12} dollars (c) $(1.2 \times 10^{12}) \div (4 \times 10^7) = 0.3 \times 10^5 = 3 \times 10^4$ dollars per person. **19.** (a) The exponents were added correctly, but $12 \geq 10$, so the answer is not in scientific notation; $12 \times 10^{11} = 1.2 \times 10^{12}$. (b) A negative exponent moves the decimal *left*, not right; $2.6 \times 10^{-4} = 0.00026$. (c) Exponents are added only when *multiplying*; to add, match the exponents first: $5 \times 10^3 + 0.3 \times 10^3 = 5.3 \times 10^3$. (d) The exponent 2 applies to the coefficient as well: $(2 \times 10^{-3})^2 = 2^2 \times 10^{-6} = 4 \times 10^{-6}$. **20.** (a) $365 \times 24 \times 60 \times 60 = 31\,536\,000$ s $= 3.1536 \times 10^7$ s. (b) $(1.4 \times 10^{10})(3.15 \times 10^7) = 4.41 \times 10^{17}$ s.

Lesson 9 — Extra Practice

Rational Exponents and Radical Form

Part A — Converting Both Ways

Throughout this set, assume every variable represents a **positive** number, so every root exists and every conversion is legal. Remember the two-way rule $x^{m/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$: the denominator is the index, the numerator is the power.

1. Write each power as an equivalent radical. Every numerator here is larger than 1.

(a) $x^{3/4}$ (b) $y^{5/2}$ (c) $m^{4/7}$ (d) $n^{7/3}$

2. Write each radical as a power with a rational exponent.

(a) $\sqrt[5]{a^2}$ (b) $\sqrt[3]{b^7}$ (c) $\sqrt[8]{c^5}$ (d) $\sqrt{d^7}$

3. Convert each expression to the other form. A negative exponent still means "reciprocal," exactly as in Lesson 6.

(a) $p^{-2/3}$ (b) $q^{-5/4}$ (c) $\dfrac{1}{\sqrt[4]{r^3}}$ (d) $\dfrac{1}{\sqrt[6]{s^5}}$

4. Write each power as a radical. Watch carefully whether the coefficient is *inside* the base or not.

(a) $5x^{1/3}$ (b) $(5x)^{1/3}$ (c) $2w^{3/5}$ (d) $(2w)^{3/5}$

Part B — Evaluating: Root First, Then Power

5. Evaluate without a calculator. Take the root first — the numbers stay small.

(a) $8^{2/3}$ (b) $16^{3/4}$ (c) $32^{2/5}$ (d) $81^{3/4}$

6. Evaluate without a calculator.

(a) $125^{4/3}$ (b) $64^{5/6}$ (c) $243^{3/5}$ (d) $36^{3/2}$

7. Evaluate without a calculator. Deal with the negative sign first: take the reciprocal, then root, then power.

(a) $16^{-3/4}$ (b) $27^{-2/3}$ (c) $32^{-3/5}$ (d) $100^{-3/2}$

8. Evaluate without a calculator. Apply the exponent to the numerator and the denominator separately.

(a) $\left(\frac{8}{125}\right)^{2/3}$ (b) $\left(\frac{16}{81}\right)^{-3/4}$ (c) $\left(\frac{4}{9}\right)^{-3/2}$ (d) $\left(\frac{27}{8}\right)^{-2/3}$

9. **Which order is easier?** Evaluate $64^{5/6}$ two different ways: first as $\left(\sqrt[6]{64}\right)^5$, and then as $\sqrt[6]{64^5}$. Show the numbers you would have to handle in each case, confirm the two answers agree, and explain in one sentence why rooting first is the sensible habit.

Part C — The Exponent Laws Still Apply

The laws from this unit do not care whether the exponent is a whole number or a fraction. Add exponents over a common denominator, subtract for division, multiply for a power of a power — and convert to radical form only at the very end, if at all. Leave every answer as a power with a **positive** exponent.

10. Simplify using the Product Law.

(a) $x^{1/2} \times x^{1/3}$ (b) $a^{2/3} \times a^{3/4}$ (c) $m^{5/6} \times m^{-1/3}$

11. Simplify using the Quotient Law.

(a) $\dfrac{x^{3/4}}{x^{1/4}}$ (b) $\dfrac{b^{1/2}}{b^{5/6}}$ (c) $\dfrac{c^{7/10}}{c^{-1/5}}$

12. Simplify using the Power of a Power and Power of a Product Laws.

(a) $\left(x^{2/3}\right)^6$ (b) $\left(16y^4\right)^{3/4}$ (c) $\left(8a^6b^{-3}\right)^{2/3}$

13. Simplify. Each of these needs two laws.

(a) $\dfrac{x^{1/2} \times x^{3/4}}{x^{1/4}}$ (b) $\left(\dfrac{a^{2/3}}{a^{1/6}}\right)^4$ (c) $\left(25m^{1/2}\right)^{1/2} \times m^{3/4}$

14. **Stretch.** Simplify, leaving your answer with positive exponents only:

$$\left(\frac{x^{1/2}y^{-1/3}}{x^{-1/4}}\right)^{12}.$$

Part D — Errors, Reversals, and Ordering

15. Jamil writes $8^{2/3} = \dfrac{8^2}{3} = \dfrac{64}{3}$. Explain exactly what he has confused, then evaluate $8^{2/3}$ correctly.

16. Priya evaluates $16^{3/4}$ by cubing first: she gets $16^3 = 4096$ and then cannot find $\sqrt[4]{4096}$ in her head. Finish her method to show it does work, then redo it the short way.

17. **Work backwards.** Find the missing exponent or base in each equation.

(a) $x^a = \sqrt[3]{x^2}$ (b) $16^n = 8$ (c) $x^{5/2} = 32$ (d) $27^n = \dfrac{1}{3}$

18. **Ordering.**

(a) Evaluate each of $4^{1/2}$, $8^{1/3}$, $16^{1/4}$, and 2^1. What do you notice? Explain the pattern by writing every base as a power of 2.

(b) Now arrange $9^{1/2}$, $27^{2/3}$, $16^{3/4}$, and $81^{1/4}$ from **least to greatest**. Note which two are equal.

19. Without a calculator, decide which is greater: $8^{2/3}$ or $4^{3/2}$. Then show that $2^{5/3}$ and $4^{5/6}$ are *equal*, by writing both with base 2.

20. *(Multiple Choice)* Which expression is **not** equivalent to $a^{5/3}$? A. $\sqrt[3]{a^5}$ B. $(\sqrt[3]{a})^5$ C. $(\sqrt[5]{a})^3$ D. $a \times a^{2/3}$

21. A cube has a volume of V cubic centimetres.

(a) Write a power in V for the edge length.

(b) Write a power in V for the total surface area of the cube.

(c) Use your formula to find the surface area when $V = 64$ and when $V = 1\,000$.

22. **Stretch.** Let $x > 0$.

(a) Expand $\left(x^{1/2} + x^{-1/2}\right)^2$ and show that it simplifies to $x + 2 + \dfrac{1}{x}$.

(b) Use part (a) to evaluate the expression when $x = 4$, then check your answer by substituting $x = 4$ into the original bracket first.

Answer Key. 1. (a) $\sqrt[4]{x^3}$ (b) $\sqrt{y^5}$ (c) $\sqrt[7]{m^4}$ (d) $\sqrt[3]{n^7}$. **2.** (a) $a^{2/5}$ (b) $b^{7/3}$ (c) $c^{5/8}$ (d) $d^{7/2}$. **3.** (a) $\dfrac{1}{\sqrt[3]{p^2}}$ (b) $\dfrac{1}{\sqrt[4]{q^5}}$ (c) $r^{-3/4}$ (d) $s^{-5/6}$. **4.** (a) $5\sqrt[3]{x}$ (b) $\sqrt[3]{5x}$ (c) $2\sqrt[5]{w^3}$ (d) $\sqrt[5]{(2w)^3} = \sqrt[5]{8w^3}$. **5.** (a) 4 (b) 8 (c) 4 (d) 27. **6.** (a) 625 (b) 32 (c) 27 (d) 216. **7.** (a) $\dfrac{1}{8}$ (b) $\dfrac{1}{9}$ (c) $\dfrac{1}{8}$ (d) $\dfrac{1}{1\,000}$. **8.** (a) $\dfrac{4}{25}$ (b) $\dfrac{27}{8}$ (c) $\dfrac{27}{8}$ (d) $\dfrac{4}{9}$. **9.** Root first: $\sqrt[6]{64} = 2$, then $2^5 = 32$. Power first: $64^5 = 1\,073\,741\,824$, and $\sqrt[6]{1\,073\,741\,824} = 32$. Both give 32, but the second route requires a ten-digit number, so rooting first keeps every step small enough to do mentally. **10.** (a) $x^{5/6}$ (b) $a^{17/12}$ (c) $m^{1/2}$. **11.** (a) $x^{1/2}$ (b) $\dfrac{1}{b^{1/3}}$ (c) $c^{9/10}$. **12.** (a) x^4 (b) $8y^3$ (c) $\dfrac{4a^4}{b^2}$. **13.** (a) $x^{1/2+3/4-1/4} = x$ (b) $\left(a^{1/2}\right)^4 = a^2$ (c) $5m^{1/4} \times m^{3/4} = 5m$. **14.** Inside: $x^{1/2+1/4}y^{-1/3} = x^{3/4}y^{-1/3}$; raising to the 12th gives $x^9y^{-4} = \dfrac{x^9}{y^4}$. **15.** He treated the denominator 3 as a divisor instead of as the index of a root. The 3 says "cube root," so $8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$, not $\dfrac{64}{3}$. **16.** Her method is valid: $16^3 = 4096$ and $\sqrt[4]{4096} = 8$, since $8^4 = 4096$. The short way is $16^{3/4} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$ — same answer, no large numbers. **17.** (a) $a = \dfrac{2}{3}$ (b) $n = \dfrac{3}{4}$ (since $16^{3/4} = 8$) (c) $x = 32^{2/5} = 4$ (d) $n = -\dfrac{1}{3}$ (since $27^{-1/3} = \dfrac{1}{3}$). **18.** (a) All four equal 2: $4^{1/2} = \left(2^2\right)^{1/2} = 2^1$, $8^{1/3} = \left(2^3\right)^{1/3} = 2^1$, $16^{1/4} = \left(2^4\right)^{1/4} = 2^1$. Each time the power of 2 in the base cancels the denominator of the exponent. (b) $9^{1/2} = 3$, $27^{2/3} = 9$, $16^{3/4} = 8$, $81^{1/4} = 3$, so $9^{1/2} = 81^{1/4} < 16^{3/4} < 27^{2/3}$; the first two are equal. **19.** $8^{2/3} = 4$ and $4^{3/2} = 8$, so $4^{3/2}$ is greater. For the second part, $2^{5/3} = 2^{5/3}$ and $4^{5/6} = \left(2^2\right)^{5/6} = 2^{10/6} = 2^{5/3}$, so they are equal. **20.** C, because $\left(\sqrt[5]{a}\right)^3 = a^{3/5}$, not $a^{5/3}$. (D checks out: $a^1 \times a^{2/3} = a^{5/3}$.) **21.** (a) edge $= V^{1/3}$ (b) $S = 6\left(V^{1/3}\right)^2 = 6V^{2/3}$ (c) $V = 64$: $S = 6(16) = 96$ cm^2; $V = 1\,000$: $S = 6(100) = 600$ cm^2. **22.** (a) $\left(x^{1/2}\right)^2 + 2\left(x^{1/2}\right)\left(x^{-1/2}\right) + \left(x^{-1/2}\right)^2 = x + 2x^0 + x^{-1} = x + 2 + \dfrac{1}{x}$. (b) At $x = 4$: $4 + 2 + \dfrac{1}{4} = \dfrac{25}{4}$; checking, $4^{1/2} + 4^{-1/2} = 2 + \dfrac{1}{2} = \dfrac{5}{2}$ and $\left(\dfrac{5}{2}\right)^2 = \dfrac{25}{4}$.

Lesson 10 — Extra Practice

Unit Review: Exponents

Part A — Multi-Law Simplification

Every question in this part needs at least two of the unit's laws. Clear the brackets first, then combine matching bases, and clean up zero, negative, and rational exponents last. Write every final answer with **positive exponents only**. Assume all variables are non-zero.

1. Simplify.

(a) $\dfrac{\left(3x^2\right)^3 \cdot x^{-4}}{9x^0}$ (b) $\dfrac{\left(2a^{-3}b^2\right)^{-2}}{8a^0b^{-5}}$ (c) $\left(\dfrac{4m^5}{m^{-1}}\right)^{-2} \times 32m^6$

2. Simplify. Here the rational exponents meet the product, quotient, and power laws.

(a) $\left(x^{1/2}\right)^4 \cdot x^{-3}$ (b) $\dfrac{y^{3/4} \cdot y^{-1/4}}{y^{-1}}$ (c) $\left(t^{-2/3}\right)^{-6} \div t^5$

3. **Stretch.** A negative *rational* exponent on the outside of a quotient — flip the fraction, then apply the root and the power to every factor.

(a) $\left(\dfrac{27x^{-3}y^6}{8x^3}\right)^{-2/3}$ (b) $\left(\dfrac{16a^8}{81b^{-4}}\right)^{-3/4}$

Part B — Evaluating: Several Laws, One Number

4. Evaluate. Collapse the powers with the laws *before* you reach for arithmetic.

(a) $\dfrac{2^5 \cdot 2^{-3}}{2^0}$ (b) $\left(\frac{8}{27}\right)^{-2/3}$ (c) $16^{-3/4}$ (d) $\dfrac{3^{-2} \cdot 3^4}{3^0 \cdot 3^3}$

5. Evaluate. Watch the negative bases and the reciprocals.

(a) $(-8)^{-2/3}$ (b) $\left(\frac{1}{4}\right)^{-3/2}$ (c) $25^{1/2} \cdot 25^{-3/2}$ (d) $\dfrac{\left(2^{-1}\right)^{-4}}{2^2 \cdot 2^0}$

6. Let $p = 16$, $q = -27$, and $r = 5$. Evaluate each expression exactly.
(a) $p^{-3/4}$ (b) $q^{2/3} + r^0$ (c) $p^{1/2} \cdot q^{-1/3}$

Part C — Scientific Notation Meets the Laws

7. Evaluate, giving each answer in **proper** scientific notation.
(a) $\dfrac{\left(4 \times 10^{6}\right)^{2}}{8 \times 10^{-3}}$ (b) $\dfrac{\left(3 \times 10^{-4}\right)^{3}}{\left(6 \times 10^{2}\right)^{-1}}$

8. Evaluate, giving each answer in proper scientific notation. In part (b), rewrite the root as a rational exponent first.
(a) $\left(2.5 \times 10^{-3}\right)^{-2}$ (b) $\sqrt{9 \times 10^{-6}}$

9. *(calculator)* A proton has a mass of about 1.67×10^{-27} kg, and an electron about 9.11×10^{-31} kg. How many times more massive is a proton than an electron? Give your answer in scientific notation, rounded to three significant digits.

Part D — Reasoning and Error Analysis

10. A student simplified $\dfrac{\left(2x^3\right)^{-2} \times 8x^5}{4x^{-1}}$ as shown below. **Exactly one step is wrong.** Identify it, explain the error, and give the correct final answer.

$$\text{Step 1: } \left(2x^3\right)^{-2} = 2^{-2}x^{-6} = \frac{1}{4x^6} \qquad \text{Step 2: } \frac{1}{4x^6} \times 8x^5 = \frac{8x^5}{4x^6} = \frac{2}{x}$$

$$\text{Step 3: } \frac{2}{x} \div 4x^{-1} = \frac{2}{x} \times \frac{1}{4x} = \frac{2}{4x^2} = \frac{1}{2x^2}$$

11. Each student below made a single mistake. Name the mistake and give the correct answer.

(a) Mira wrote $16^{3/4} = \sqrt[3]{16^4}$.

(b) Owen wrote $3x^{-2} = \dfrac{1}{3x^2}$.

12. Decide whether each statement is **true** or **false**. If it is false, give a counterexample *and* state the correct version.

(a) $(a+b)^2 = a^2 + b^2$ for all values of a and b.

(b) a^{-1} means the opposite of a.

(c) $\left(a^2b^3\right)^0 = 1$ for *all* values of a and b.

(d) $a^{1/2} \cdot a^{1/2} = a$ for every $a \geq 0$.

Part E — Working Backwards

13. Determine the value of the unknown exponent that makes each statement true.

(a) $\dfrac{x^7 \cdot x^n}{x^{-2}} = x^4$

(b) $(2^m)^{-3} = \dfrac{1}{64}$

(c) $8^k = \dfrac{1}{4}$

14. Determine the value of the unknown that makes each statement true.

(a) $\left(\dfrac{x^a}{x^3}\right)^2 = x^{-4}$

(b) $(3 \times 10^n)^2 = 9 \times 10^{-6}$

(c) $81^{\,p} = 27$

Part F — Applications

15. An investment grows according to $P = 2000(1.05)^t$, where P is the value in dollars and t is the time in years since the investment was made.

(a) Find P when $t = 0$, and name the exponent law that makes the calculation immediate.

(b) *(calculator)* Write an expression, using a negative exponent, for the value of the investment 3 years *before* it was made, and evaluate it to the nearest cent.

16. A cube has an edge length of $2x^{-1}y^2$ centimetres.

(a) Determine the volume of the cube. Write the answer with positive exponents.

(b) The edge length is doubled. By what factor does the volume change? Justify your answer with an exponent law rather than by substituting numbers.

17. Compare each pair of measurements. Show the division and give each answer in proper scientific notation.

(a) A red blood cell is about 7×10^{-6} m wide; a water molecule is about 2.8×10^{-10} m wide. How many times wider is the cell?

(b) The Sun has a mass of about 2×10^{30} kg and Earth about 6×10^{24} kg. How many Earths would it take to match the Sun's mass?

Part G — Challenge

18. Without a calculator, decide which is larger: 2^{100} or 10^{30}. *Hint:* rewrite 2^{100} as a power of 2^{10}, and compare that base with 10^3. Explain your reasoning in a sentence.

19. Simplify completely. Write each answer with positive exponents.

(a) $\left(\dfrac{x^{-1/2}y^2}{x^{3/2}y^{-2}}\right)^{-3} \times x^{-6}$

(b) $\dfrac{\left(8a^6\right)^{2/3} \cdot a^{-5}}{\left(4a^{-2}\right)^{-1/2}}$

20. Compare without a calculator, writing every power with the same base first.

(a) Which is greater, $\left(\frac{1}{2}\right)^{-10}$ or 4^6? By what factor?

(b) Arrange from least to greatest: $9^{-1/2}$, $27^{-2/3}$, $\left(\frac{1}{81}\right)^{1/4}$. Note carefully if any two are equal.

Answer Key. 1. (a) $\frac{27x^6 \cdot x^{-4}}{9} = 3x^2$ (b) $\frac{2^{-2}a^6b^{-4}}{8b^{-5}} = \frac{a^6b^5}{4 \cdot 8\, b^4} = \frac{a^6 b}{32}$ (c) inside $= 4m^6$, so $\left(4m^6\right)^{-2} \times 32m^6 = \frac{32m^6}{16m^{12}} = \frac{2}{m^6}$. **2.** (a) $x^2 \cdot x^{-3} = x^{-1} = \frac{1}{x}$ (b) $\frac{y^{1/2}}{y^{-1}} = y^{3/2}$ (c) $\frac{t^4}{t^5} = t^{-1} = \frac{1}{t}$. **3.** (a) inside $= \frac{27y^6}{8x^6}$; flipping gives $\left(\frac{8x^6}{27y^6}\right)^{2/3} = \frac{4x^4}{9y^4}$ (b) inside $= \frac{16a^8b^4}{81}$; flipping gives $\left(\frac{81}{16a^8b^4}\right)^{3/4} = \frac{27}{8a^6b^3}$. **4.** (a) $2^{5-3-0} = 2^2 = 4$ (b) $\left(\frac{27}{8}\right)^{2/3} = \frac{9}{4}$ (c) $\frac{1}{\left(\sqrt[4]{16}\right)^3} = \frac{1}{8}$ (d) $3^{-2+4-0-3} = 3^{-1} = \frac{1}{3}$. **5.** (a) $\frac{1}{\left(\sqrt[3]{-8}\right)^2} = \frac{1}{(-2)^2} = \frac{1}{4}$ (b) $4^{3/2} = 8$ (c) $25^{1/2-3/2} = 25^{-1} = \frac{1}{25}$ (d) $\frac{2^4}{2^2} = 4$. **6.** (a) $16^{-3/4} = \frac{1}{8}$ (b) $(-27)^{2/3} = (-3)^2 = 9$, and $5^0 = 1$, so 10 (c) $4 \times \frac{1}{-3} = -\frac{4}{3}$. **7.** (a) $\frac{16 \times 10^{12}}{8 \times 10^{-3}} = 2 \times 10^{15}$ (b) $27 \times 10^{-12} \times 6 \times 10^2 = 162 \times 10^{-10} = 1.62 \times 10^{-8}$. **8.** (a) $\frac{1}{6.25 \times 10^{-6}} = 0.16 \times 10^6 = 1.6 \times 10^5$ (b) $\left(9 \times 10^{-6}\right)^{1/2} = 3 \times 10^{-3}$. **9.** $\frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}} = 0.183 \times 10^4 \approx 1.83 \times 10^3$ times (≈ 1830). **10.** Step 3 is wrong: dividing by $4x^{-1}$ means multiplying by $\frac{1}{4}x^1$, not $\frac{1}{4x}$ — the negative exponent is on x, so it moves *up*, not down. Correctly, $\frac{2}{x} \times \frac{x}{4} = \frac{1}{2}$. **11.** (a) She swapped the index and the power: the denominator is the index, so $16^{3/4} = \sqrt[4]{16^3} = 2^3 = 8$. (b) The exponent belongs only to x, not to the coefficient 3, so $3x^{-2} = \frac{3}{x^2}$. **12.** (a) False; with $a = b = 1$, $(1+1)^2 = 4$ but $1^2 + 1^2 = 2$. Correctly $(a+b)^2 = a^2 + 2ab + b^2$. (b) False; $2^{-1} = \frac{1}{2}$, not -2. A negative exponent means the *reciprocal*, $a^{-1} = \frac{1}{a}$. (c) False; it fails when $a = 0$ or $b = 0$, since 0^0 is undefined. It is 1 for all *non-zero* a and b. (d) True; $a^{1/2} \cdot a^{1/2} = a^{1/2+1/2} = a^1 = a$. **13.** (a) $7 + n + 2 = 4$, so $n = -5$ (b) $2^{-3m} = 2^{-6}$, so $m = 2$ (c) $2^{3k} = 2^{-2}$, so $k = -\frac{2}{3}$. **14.** (a) $2(a-3) = -4$, so $a = 1$ (b) $9 \times 10^{2n} = 9 \times 10^{-6}$, so $n = -3$ (c) $3^{4p} = 3^3$, so $p = \frac{3}{4}$. **15.** (a) $P = 2000(1.05)^0 = 2000(1) = \$2000$, by the zero-exponent law. (b) $P = 2000(1.05)^{-3} = \frac{2000}{1.157625} = \$1727.68$. **16.** (a) $V = \left(2x^{-1}y^2\right)^3 = 8x^{-3}y^6 = \frac{8y^6}{x^3}$ cm^3 (b) $(2 \cdot \text{edge})^3 = 2^3 \cdot \text{edge}^3$ by the power-of-a-product law, so the volume becomes 8 times as large. **17.** (a) $\frac{7 \times 10^{-6}}{2.8 \times 10^{-10}} = 2.5 \times 10^4$ times wider (b) $\frac{2 \times 10^{30}}{6 \times 10^{24}} = 0.33\overline{3} \times 10^6 \approx 3.3 \times 10^5$ Earths. **18.** $2^{100} = \left(2^{10}\right)^{10} = 1024^{10}$ and $10^{30} = \left(10^3\right)^{10} = 1000^{10}$. Since $1024 > 1000$ and both are raised to the same power, 2^{100} is larger. **19.** (a) inside $= \frac{y^4}{x^2}$, so $\left(\frac{y^4}{x^2}\right)^{-3} = x^6y^{-12}$, and $x^6y^{-12} \times x^{-6} = \frac{1}{y^{12}}$ (b) numerator $= 4a^4 \cdot a^{-5} = \frac{4}{a}$, denominator $= \frac{a}{2}$, so $\frac{4}{a} \times \frac{2}{a} = \frac{8}{a^2}$. **20.** (a) $\left(\frac{1}{2}\right)^{-10} = 2^{10} = 1024$ and $4^6 = 2^{12} = 4096$, so 4^6 is greater, by a factor of $2^2 = 4$. (b) $9^{-1/2} = \frac{1}{3}$, $27^{-2/3} = \frac{1}{9}$, and $\left(\frac{1}{81}\right)^{1/4} = \frac{1}{3}$, so $27^{-2/3} < 9^{-1/2} = \left(\frac{1}{81}\right)^{1/4}$ — the first and third are equal.